SI-2026

Analog IC Design

Introduction to CMOS VLSI Design

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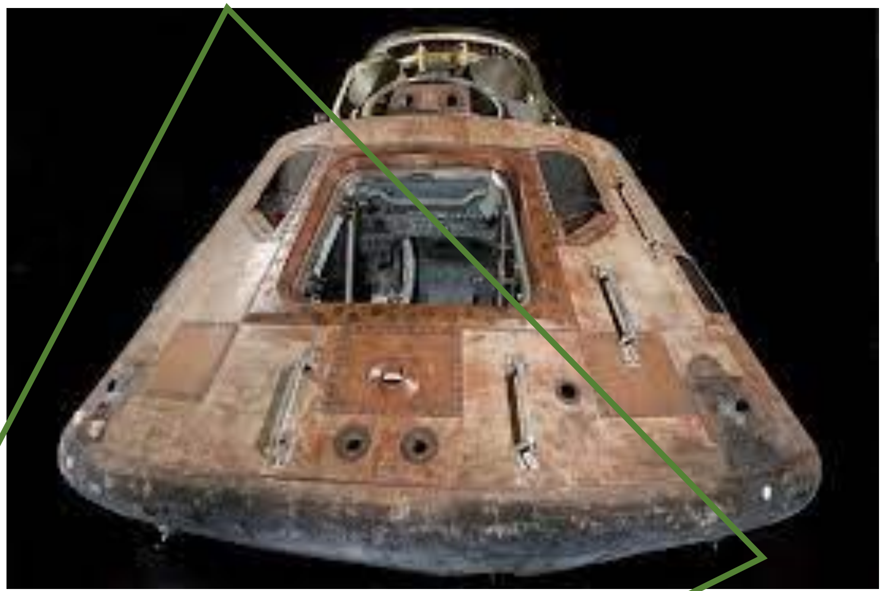
Summer Internship

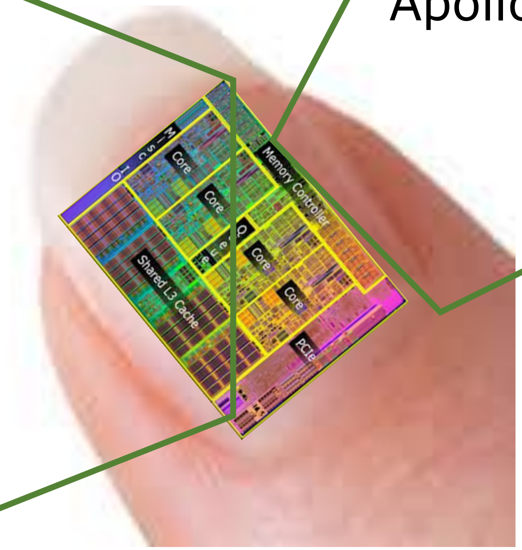
June 02, 2026

Very Large Scale Integration (VLSI)



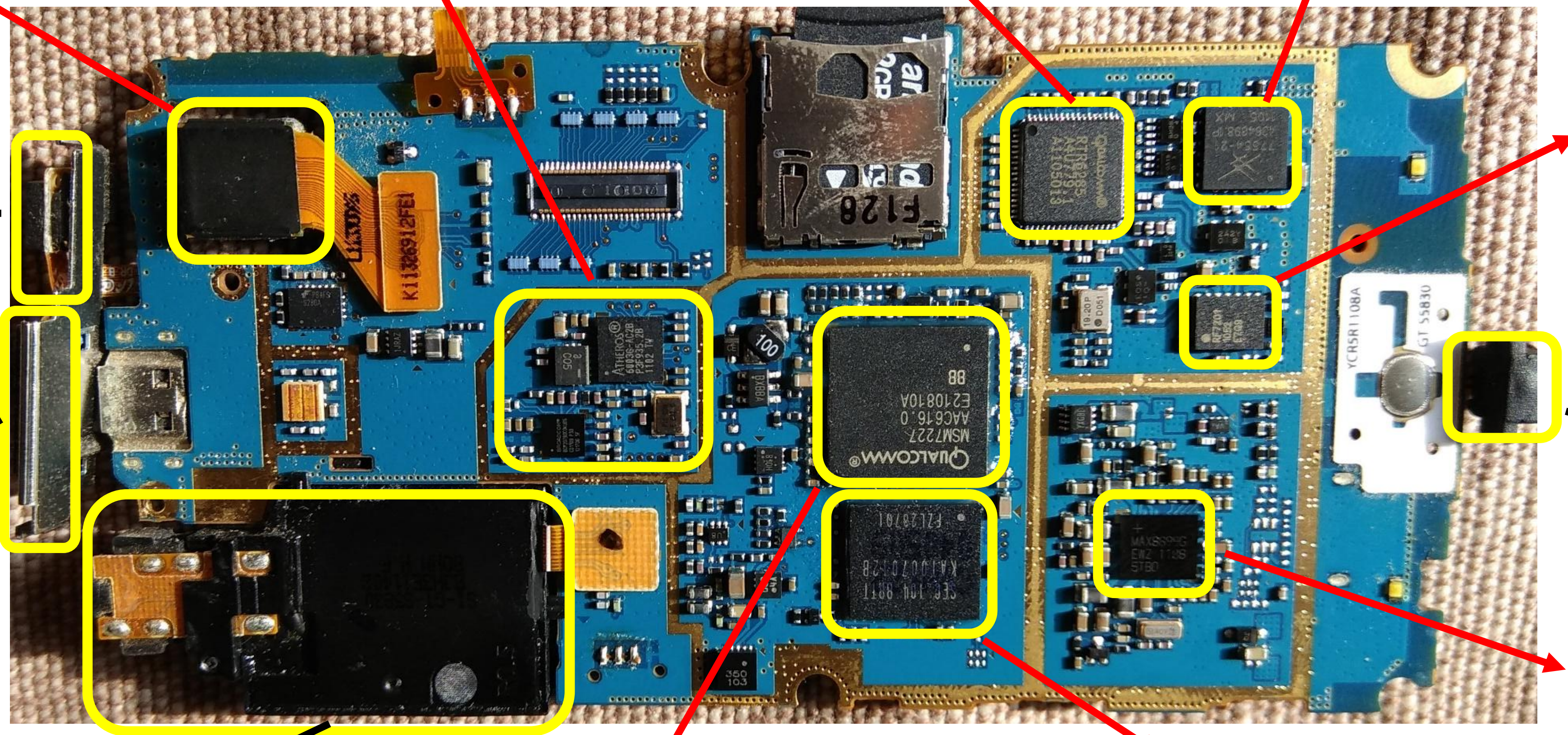
Apollo 11 Lunar Command Module 1969

Mainframe computer of the 70s



Today

Anatomy of an Electronic System: Mobile Phone



CMOS Imager

WiFi+Bluetooth Transciever

GSM/GPRS /GPS CMOS Transceiver

GSM/GPRS /GPS CMOS Front-end Chip

GaAs Power Amplifier

Proximity Sensor

Ear phone

MEMS+CMOS microphone

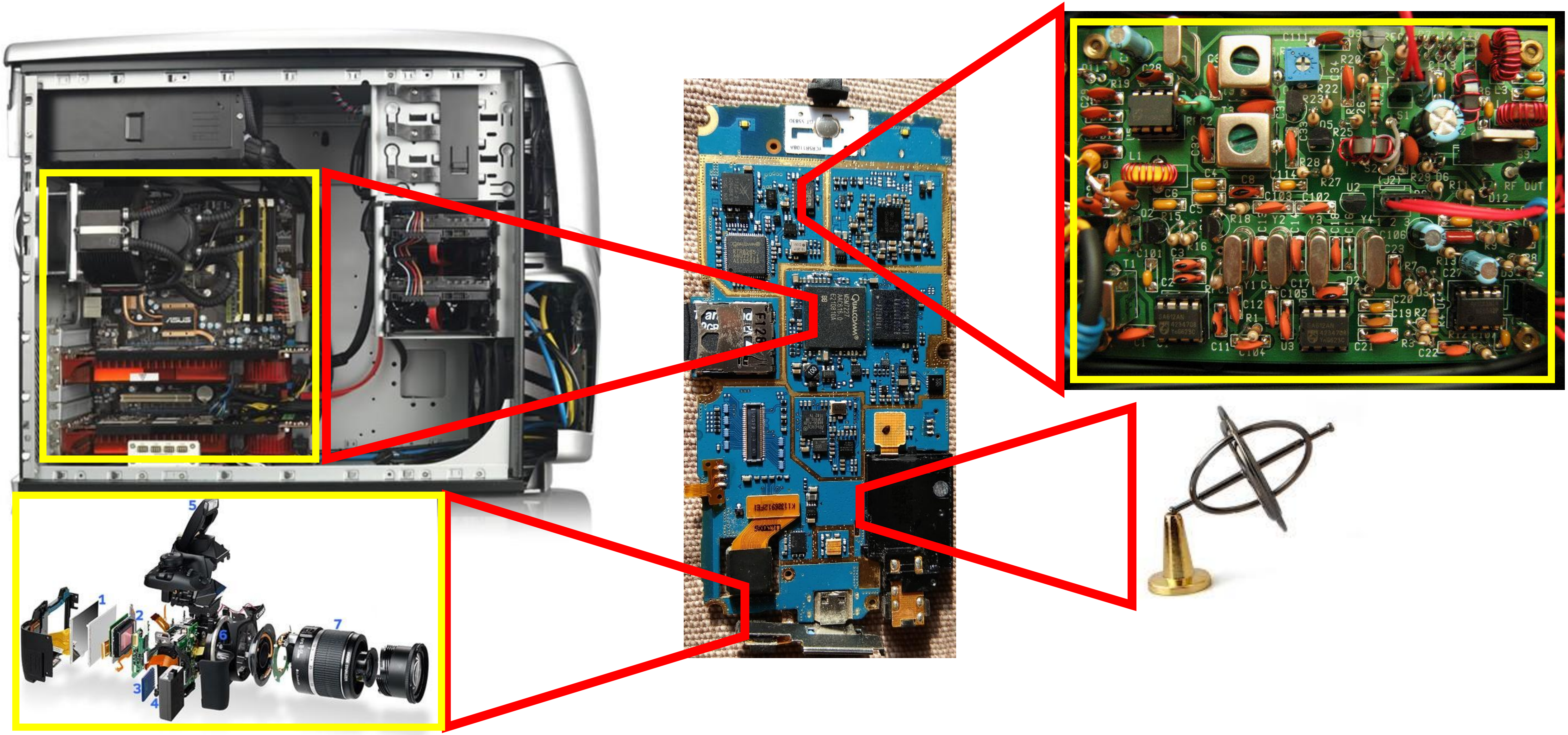
CMOS Power Management

Speaker Headphone driver

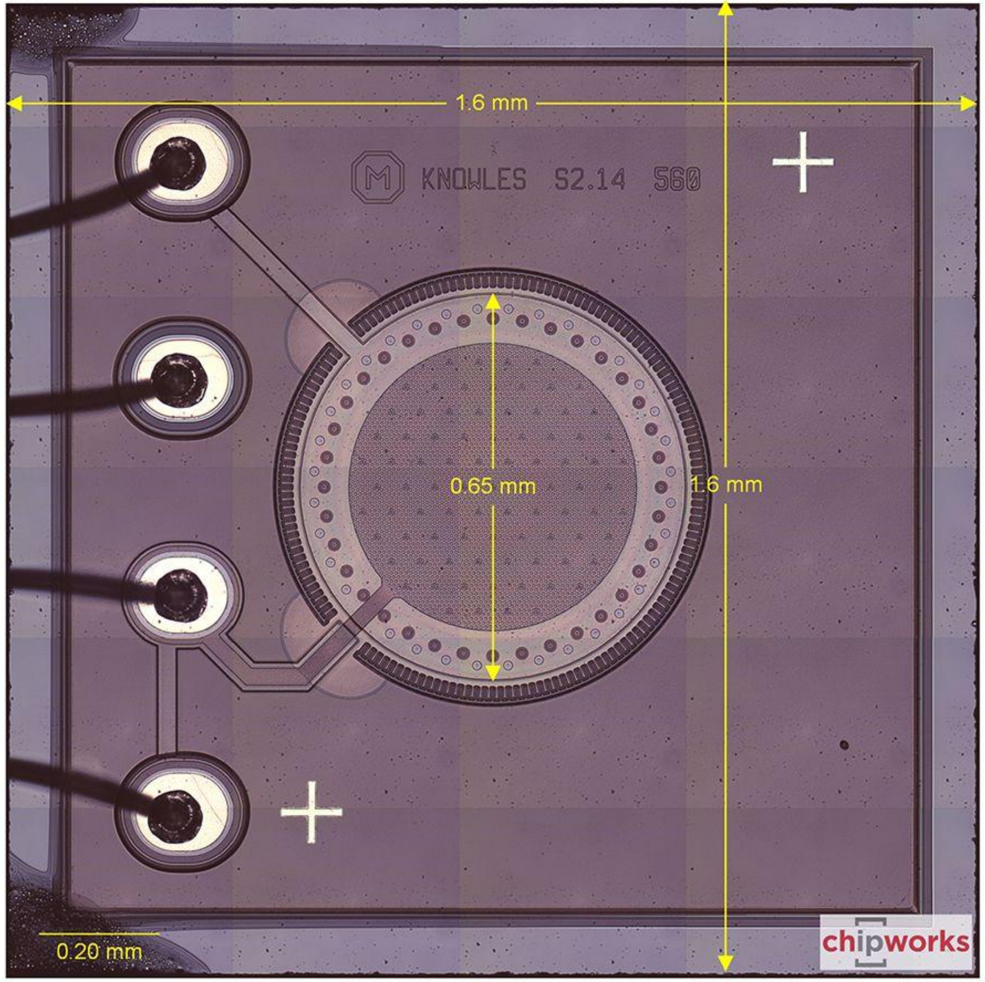
Qualcomm Snapdragon S1
1 GHz ARM Processor
CMOS Technology

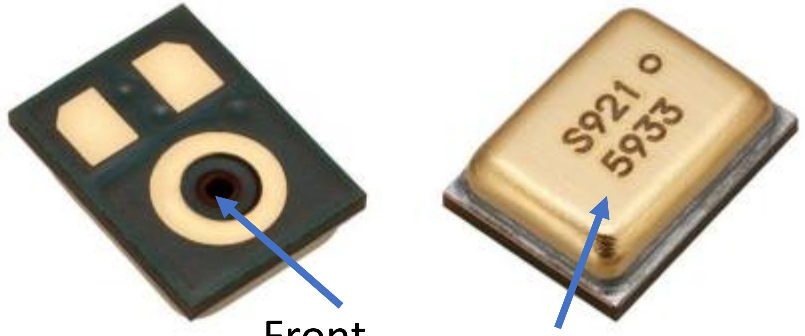
CMOS Memory

Very Large Scale Integration (VLSI)



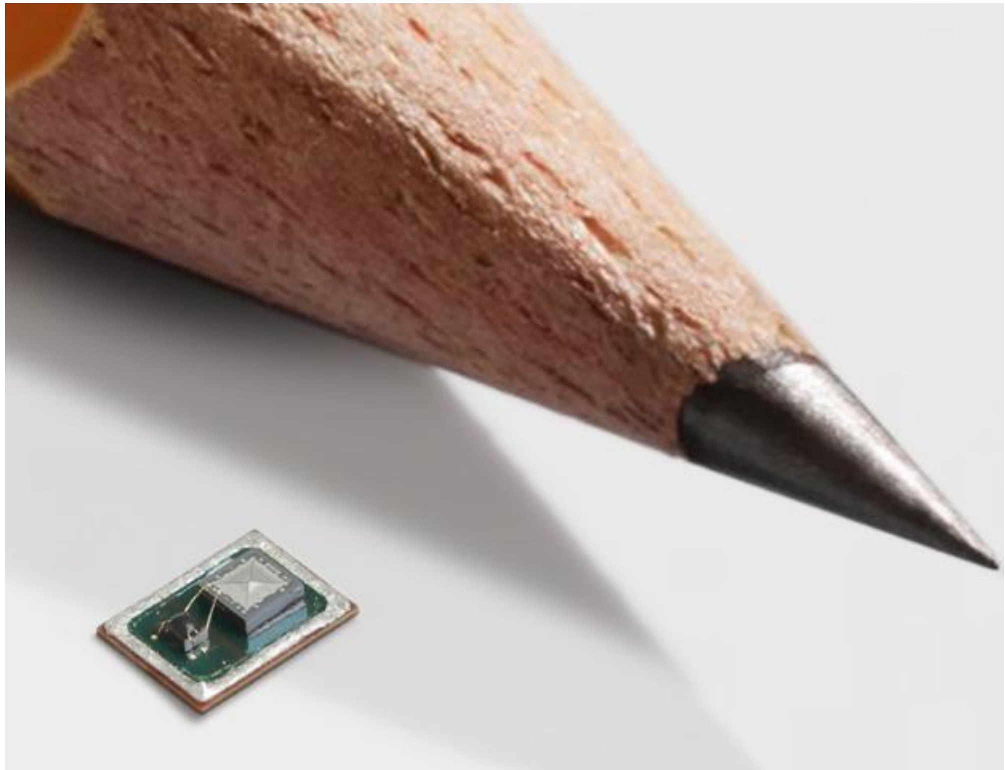
Capacitive MEMS Microphone

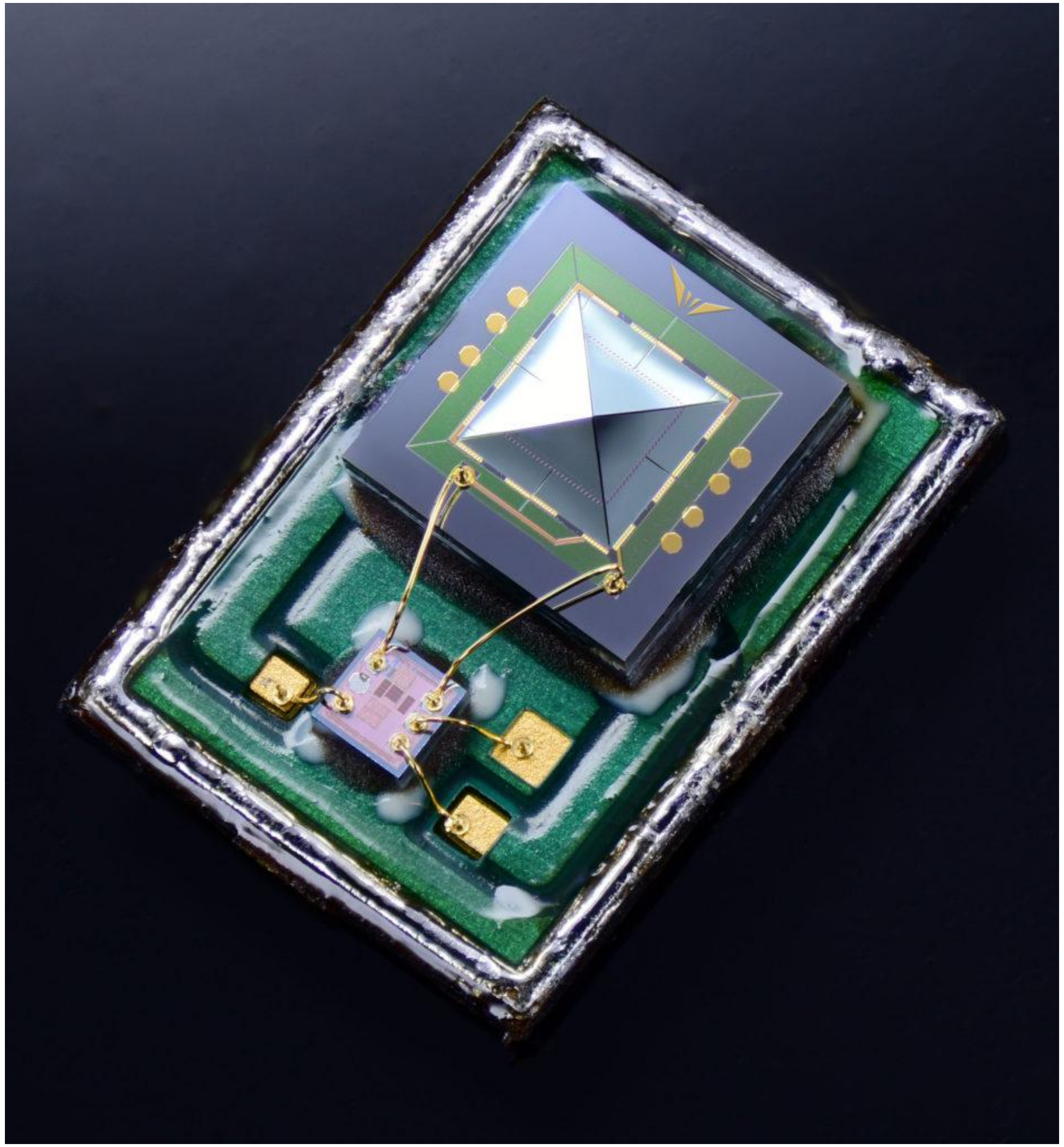




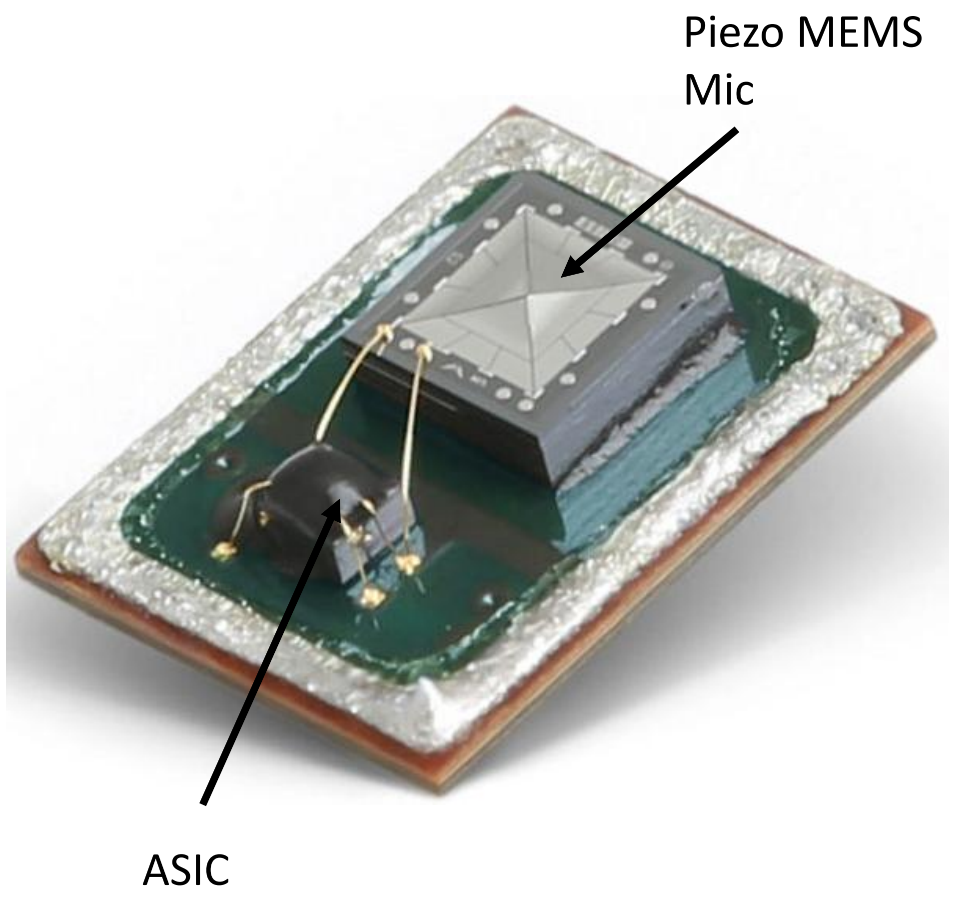
Front Cavity

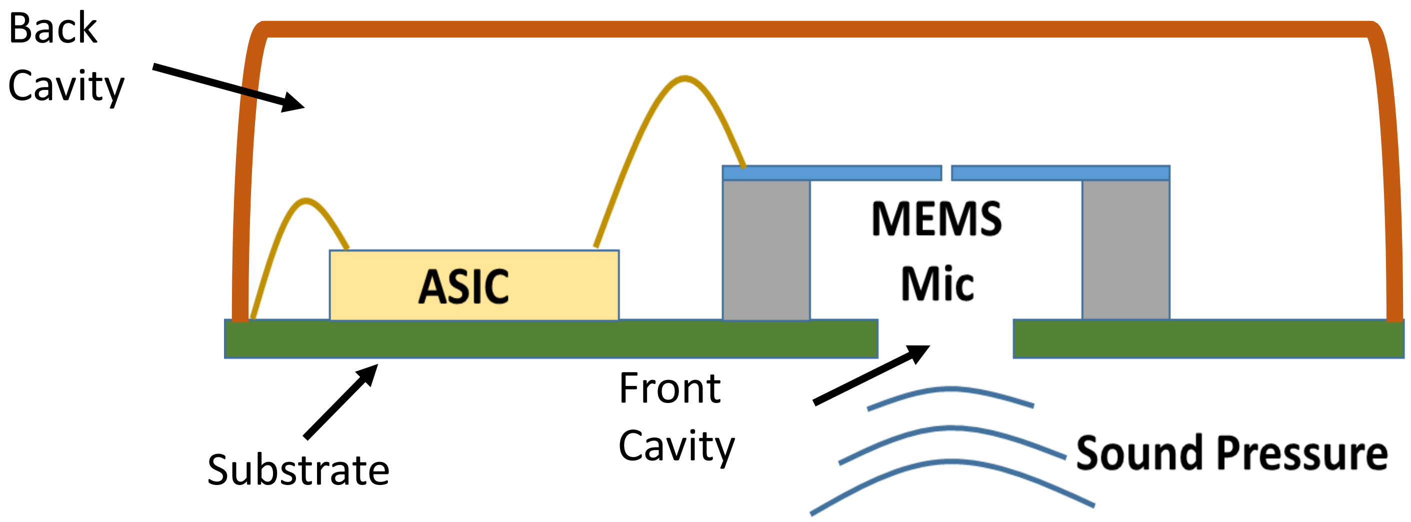
Piezoelectric MEMS Microphone



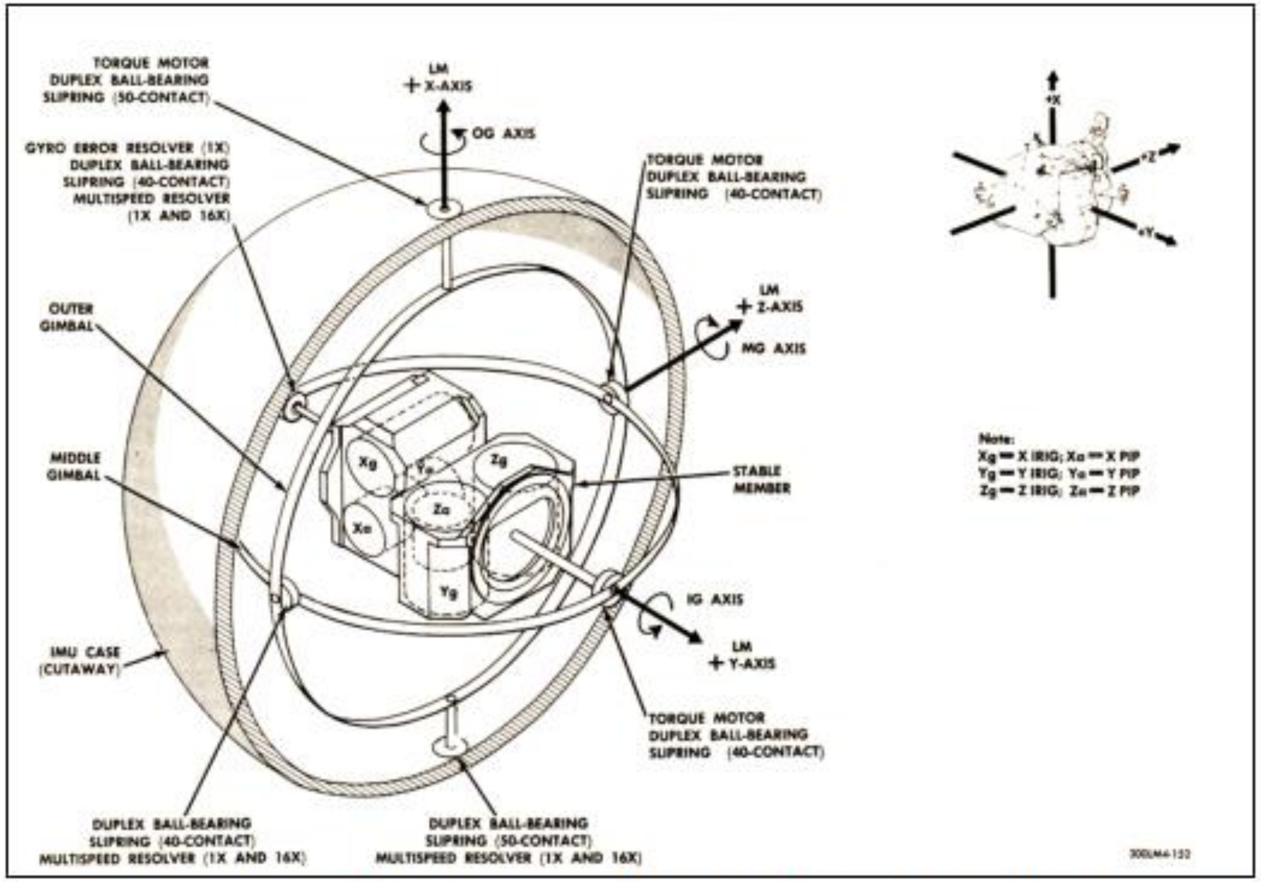


Cross-section





Inertial Measurement Unit (IMU)/ Gyroscope



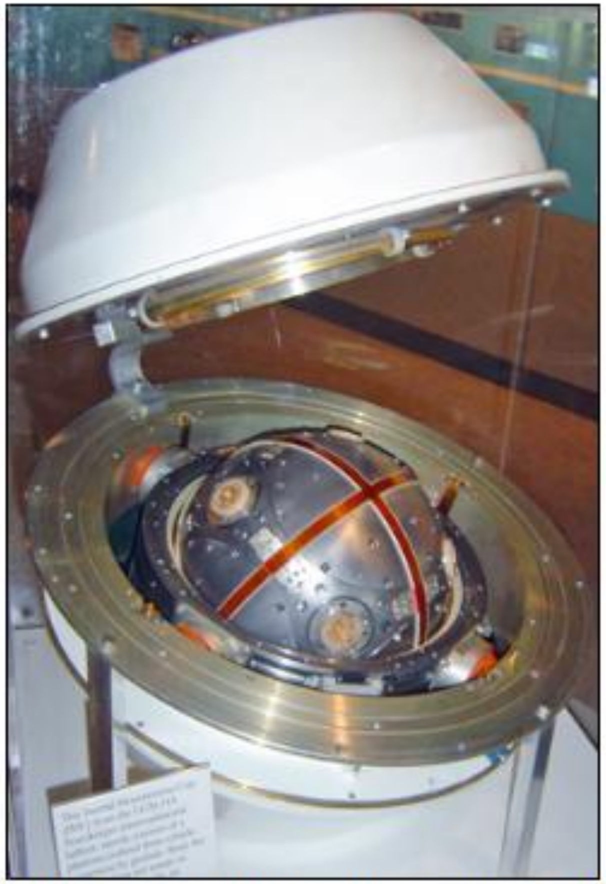


Figure 1.2: (left) Platform-type IMU utilized in the lunar module flight control during the Apollo missions. Three orthogonal accelerometers and three orthogonal gyroscopes are placed in a platform stabilized by a gimbal structure [10]. (right) IMU utilized in the LGM-118 peacemaker intercontinental ballistic missile [11].

MEMS/CMOS Gyroscope

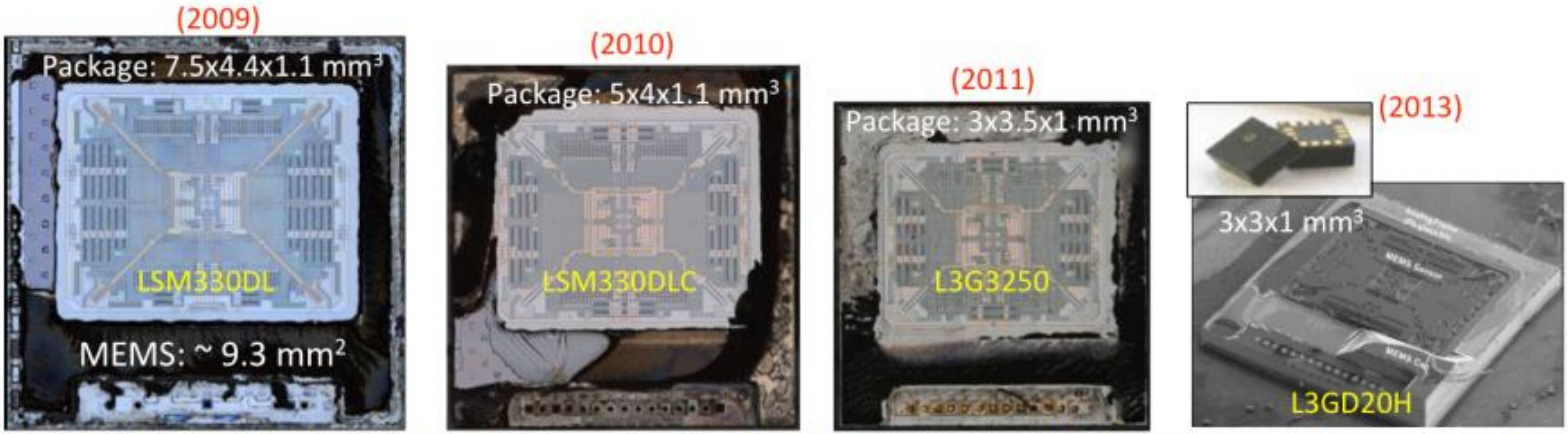
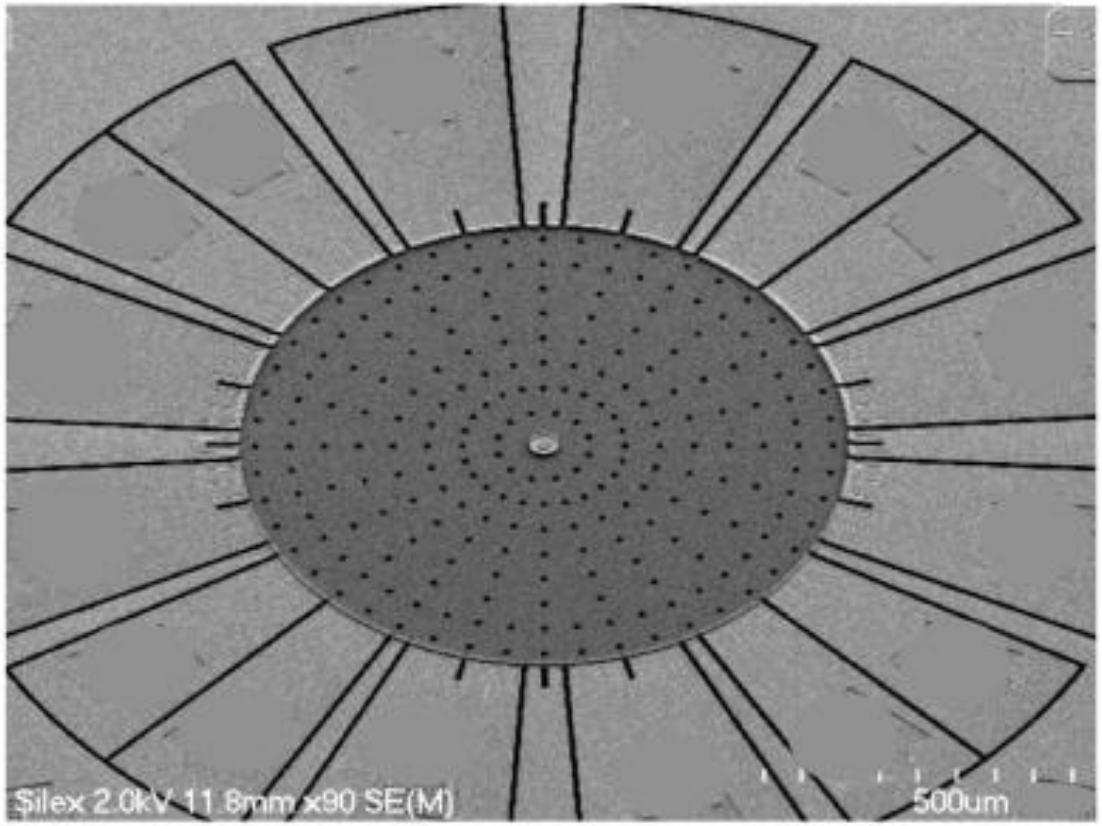


Figure 1.6: Evolution of STmicroelectronics tri-axial gyroscopes. Package size volume reduced by 4X in 4 years.

MEMS/CMOS Gyroscope



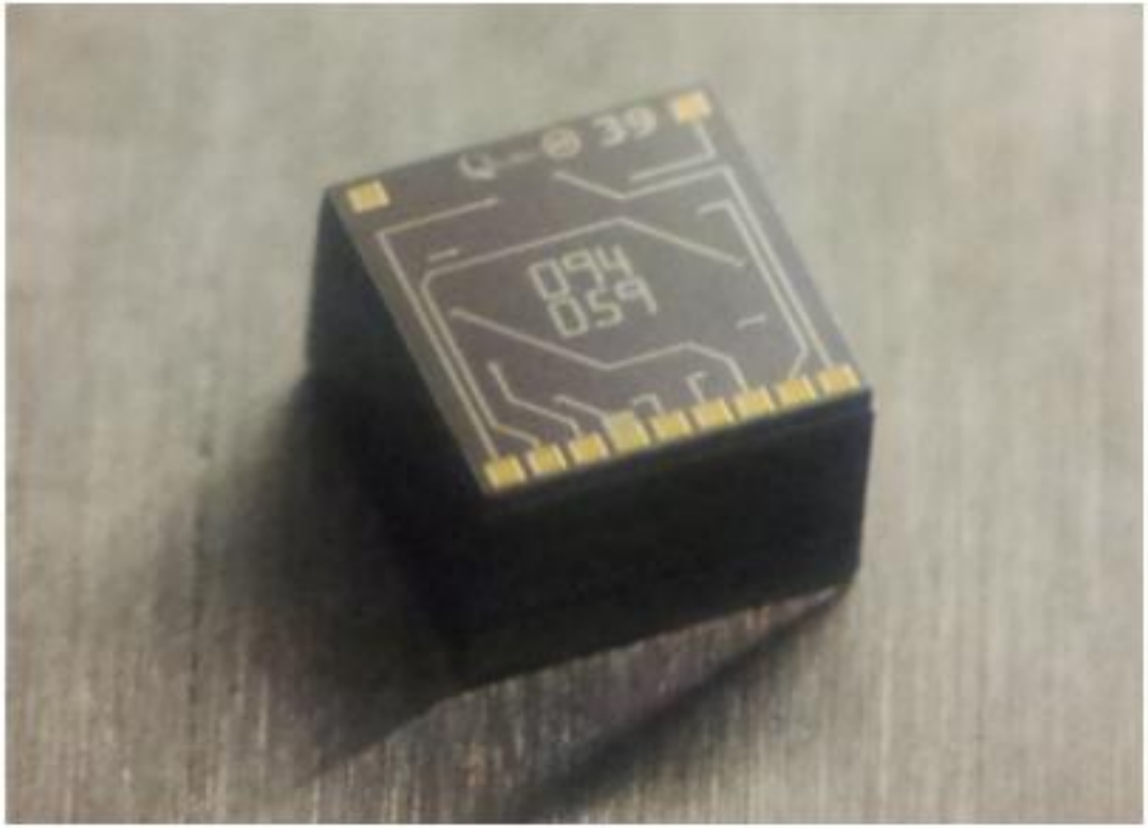
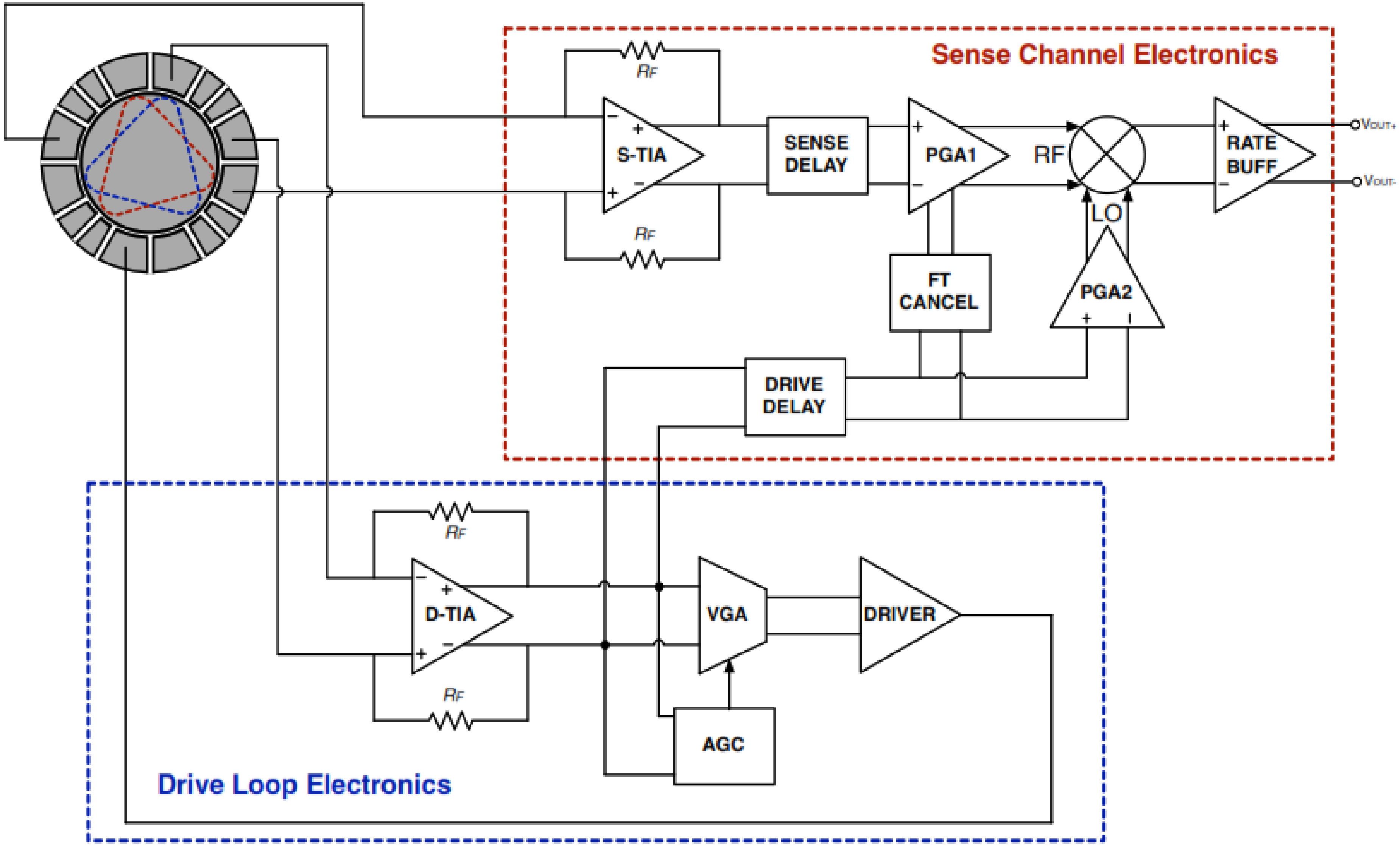
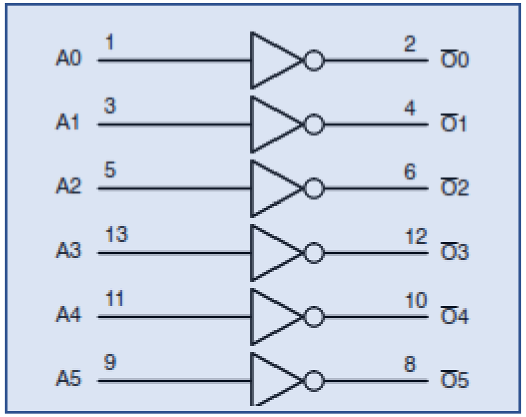


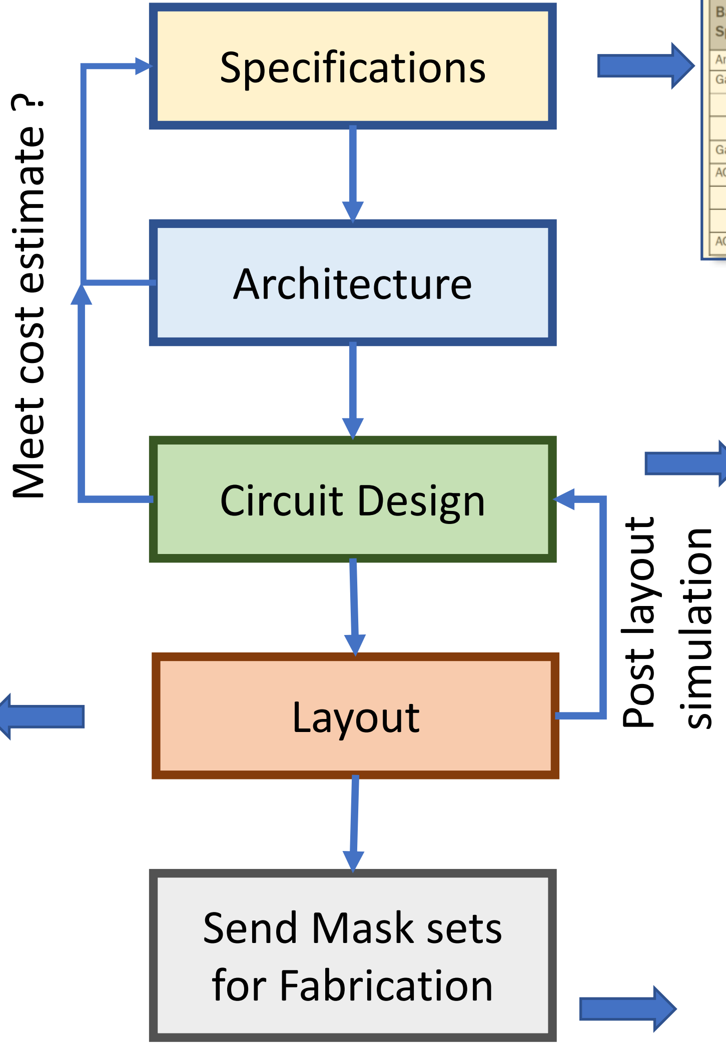
Figure 4.22: (left) SEM view of 800 µm BAW disk gyroscope implemented with the HARPSS™ process in a 40 µm-thick substrate. (right) Vacuum-packaged BAW disk gyroscope singulated die.

Analog Front-End (AFE)

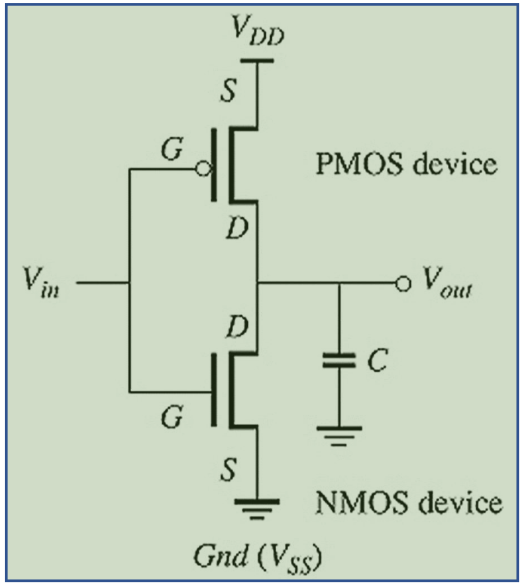


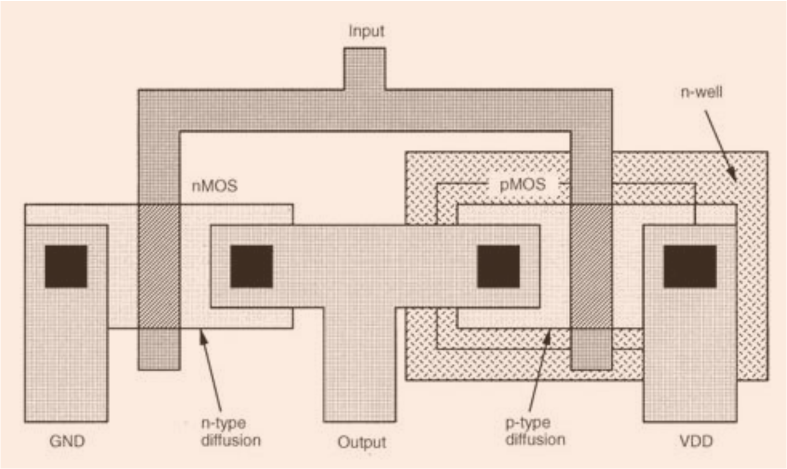
▸ *Custom (Analog/Digital)* Integrated Circuit (IC) Design Flow

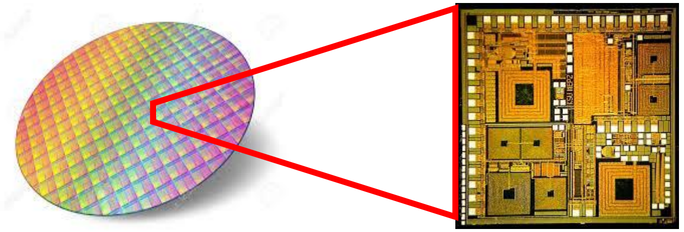




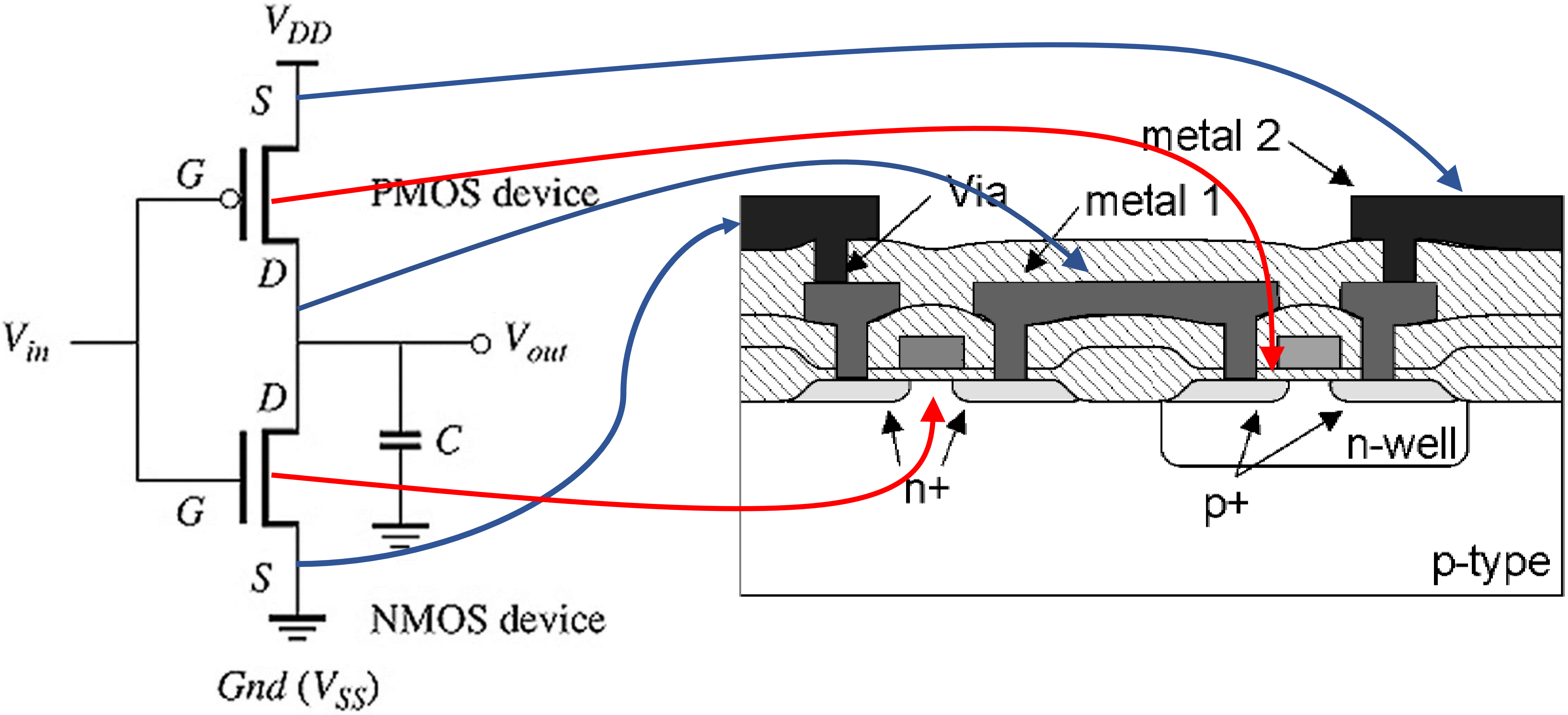
Parameter	Specification			Unit	Condition
	Min.	Typ.	Max.		
Band 1 Electrical Specifications					$T=+25°C$, $V_{CC}=V_{BAT}=+3.4V$, $V_{EN}=+1.8V$, Rel 99 Modulation, and 50Ω system, unless otherwise specified.
Ambient Temperature	-30	+25	+85	°C	
Gain	25	27		dB	HPM, $P_{OUT}=28.0dBm$
	15	19		dB	MPM, $P_{OUT}\leq 19.0dBm$
	13[1]	16		dB	LPM, $P_{OUT}\leq 8.0dBm$
Gain Linearity		±1.0		dB	HPM, $19.0dBm\leq P_{OUT}\leq 28.0dBm$
ACLR - 5MHz Offset		-40		dBc	HPM, $P_{OUT}=28.0dBm$
		-42		dBc	MPM, $P_{OUT}=19.0dBm$
		-42		dBc	LPM, $P_{OUT}=8.0dBm$
ACLR - 10MHz Offset		-53		dBc	HPM, $P_{OUT}=28.0dBm$



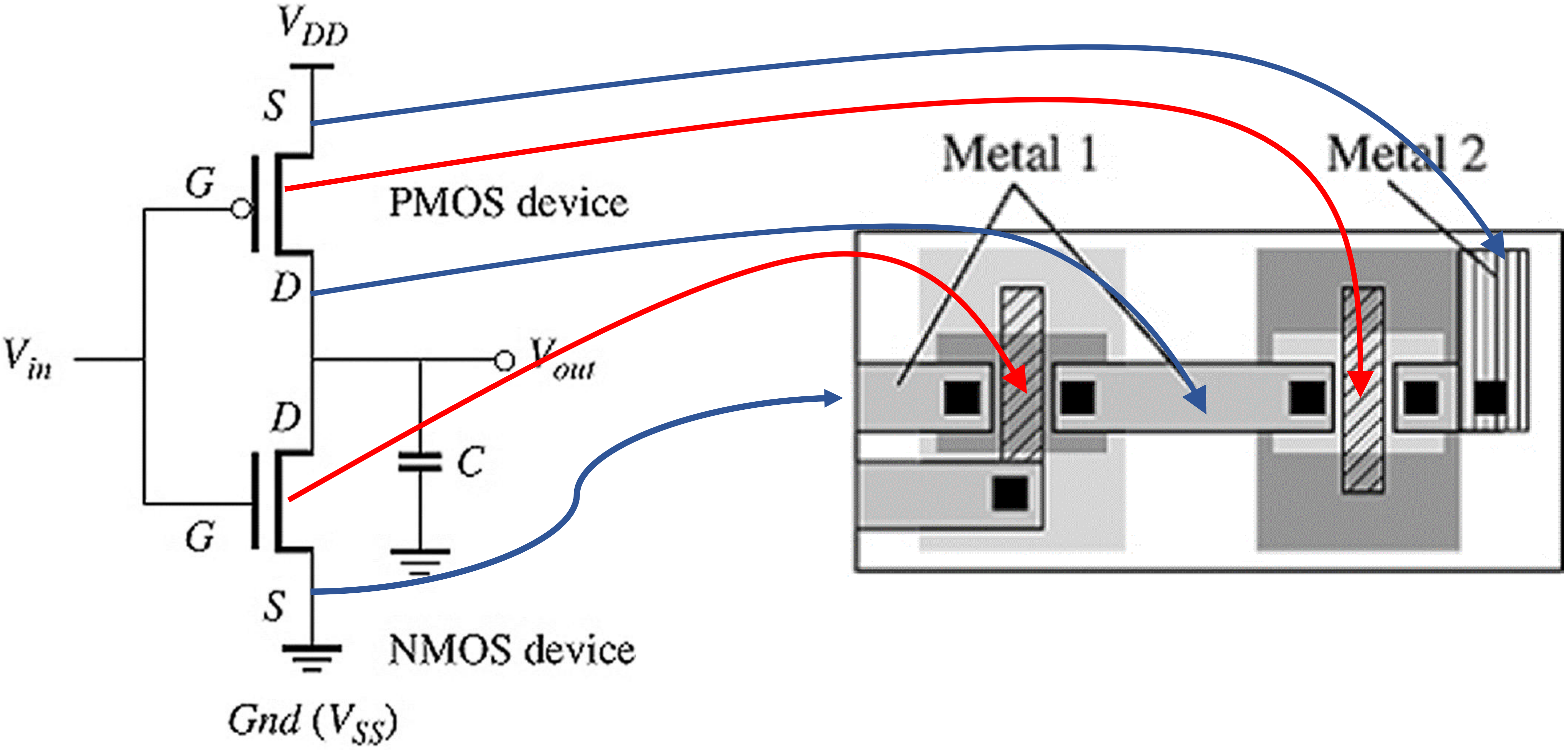




Schematic to Physical Devices

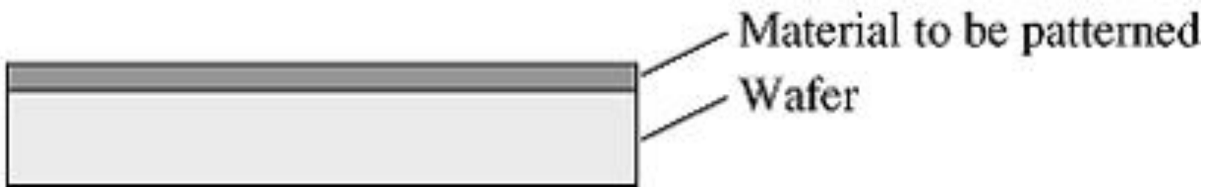


Circuit to Foundry Masks (Layout)

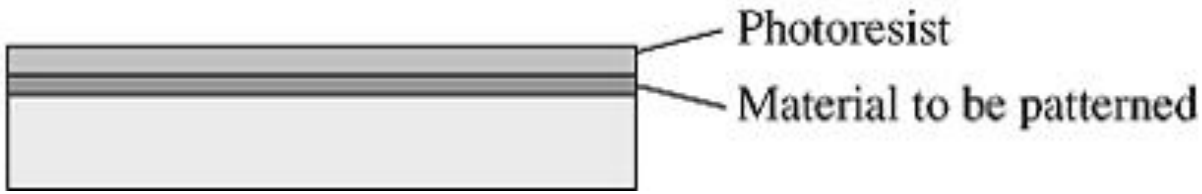


Material Patterning Steps (1)

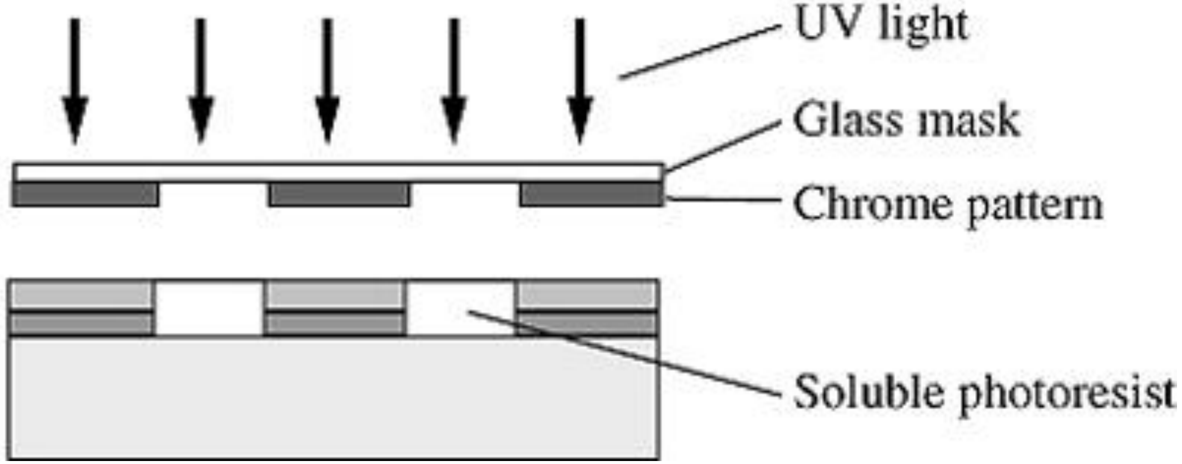
Step 1: Apply material to water



Step 2: *Spin* on a photoresistive material



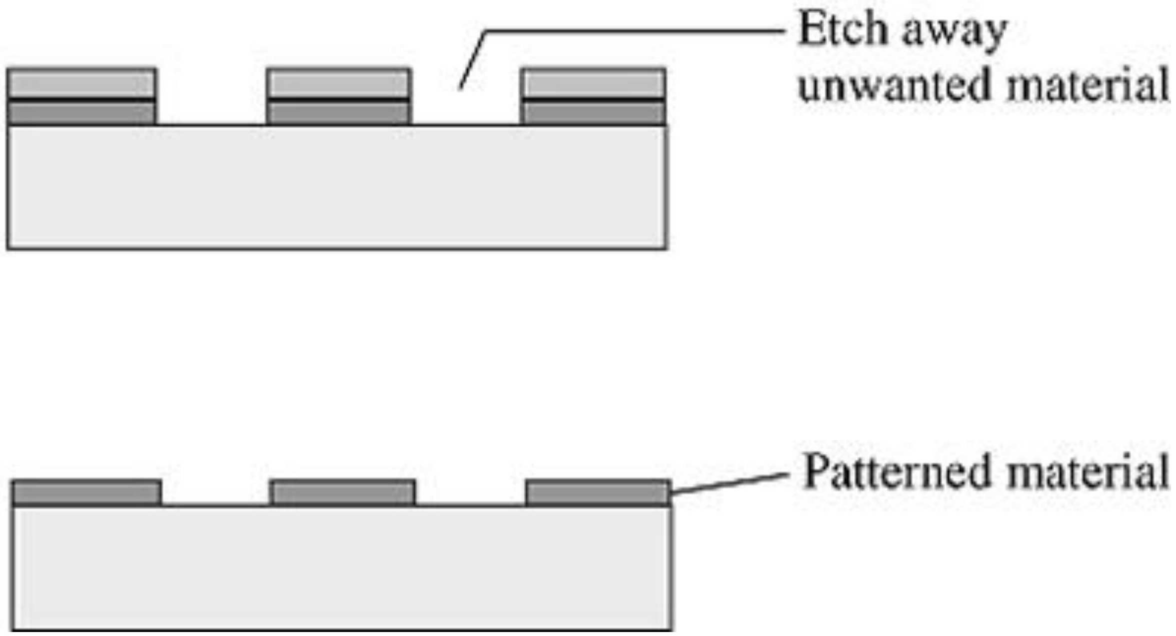
Step 3: Pattern photoresist with UV light through glass mask



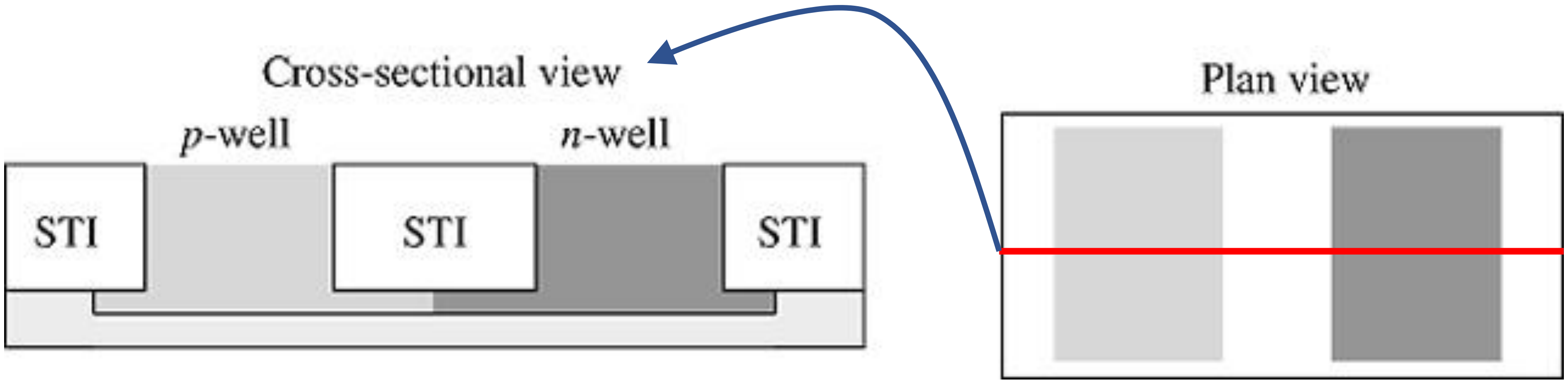
Material Patterning Steps (2)

Step 4: Apply specific processing step such as etch, implant, oxidation, after removing soluble photoresist

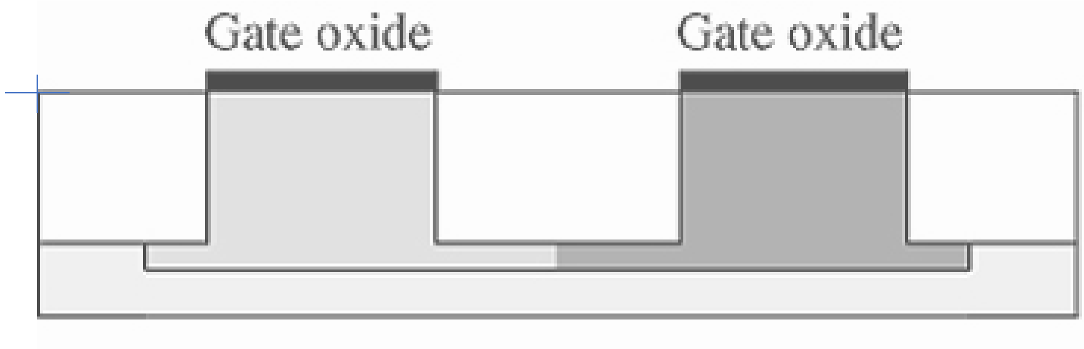
Step 5: Wash off resist

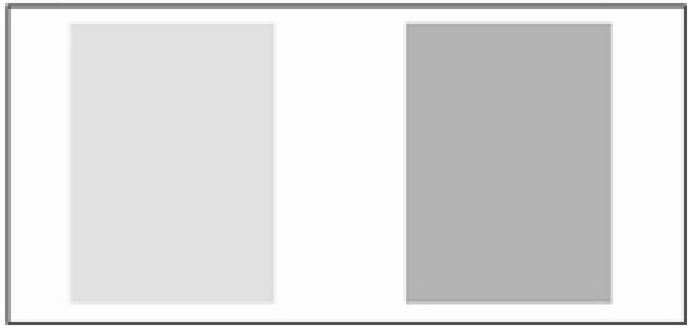


Foundry Masks (Layout) to Physical Devices

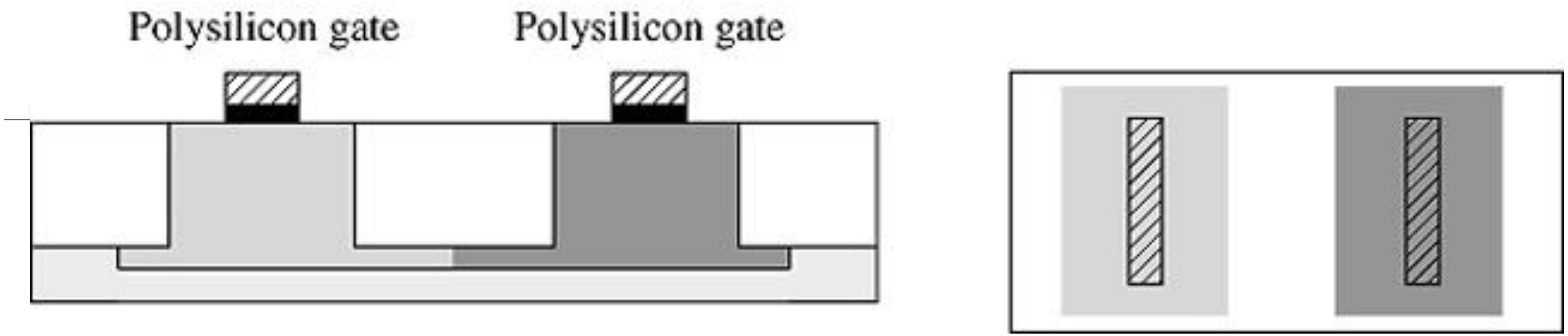


Foundry Masks (Layout) to Physical Devices (2)

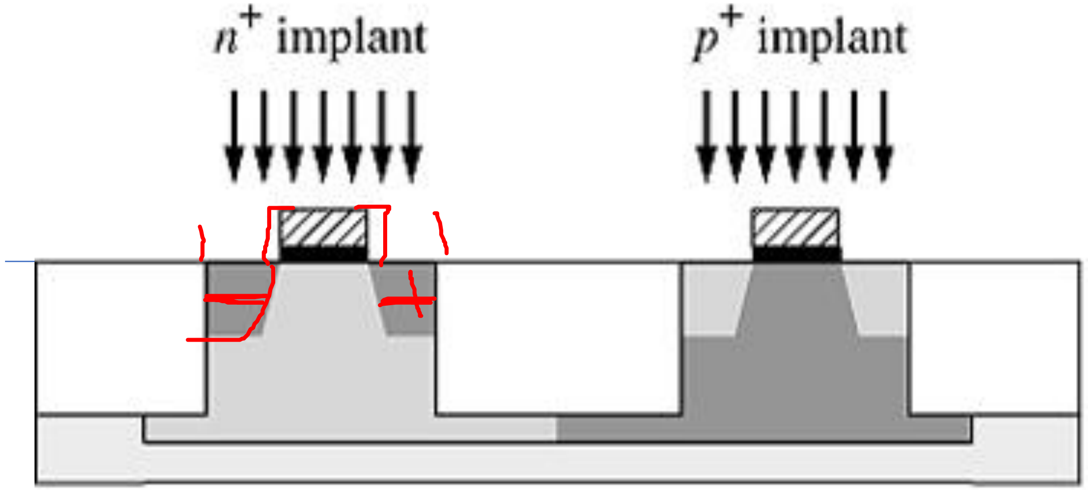


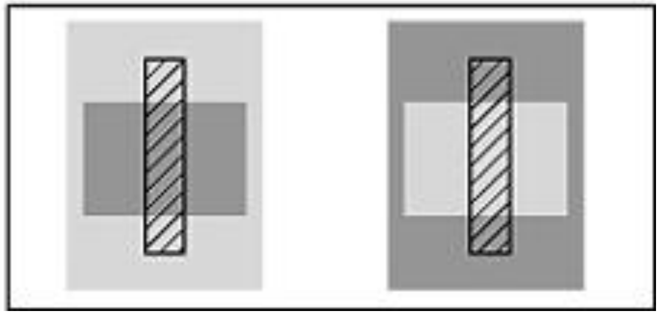


Foundry Masks (Layout) to Physical Devices (3)

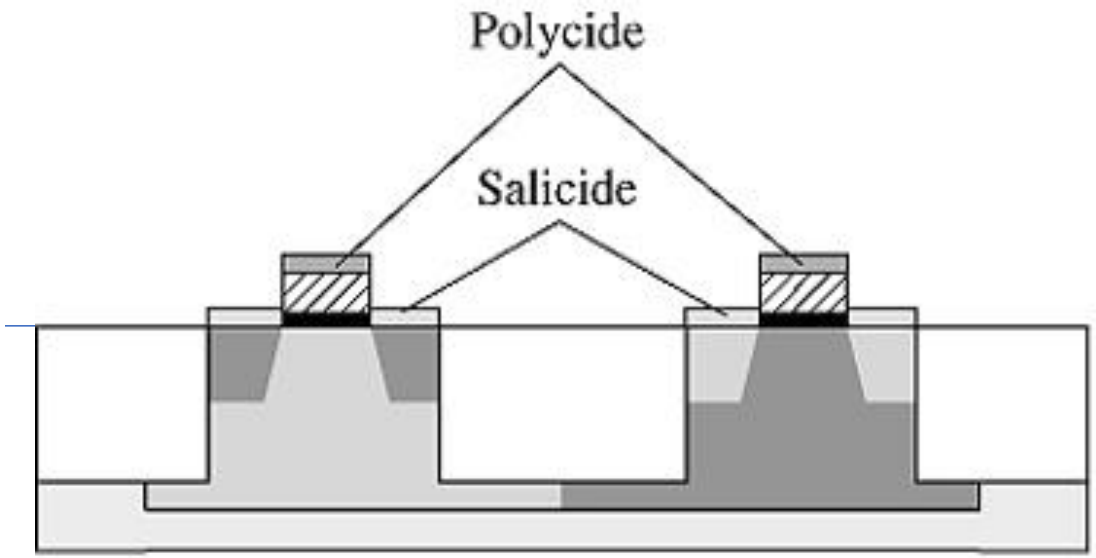


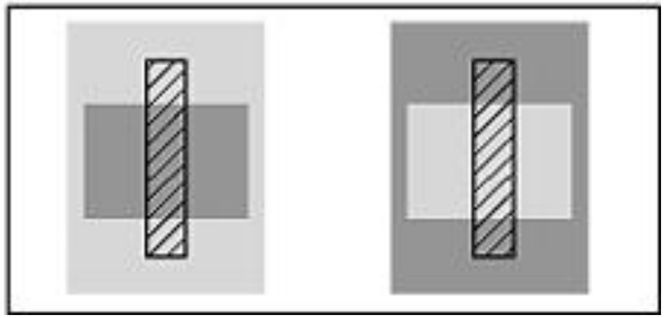
Foundry Masks (Layout) to Physical Devices (4)



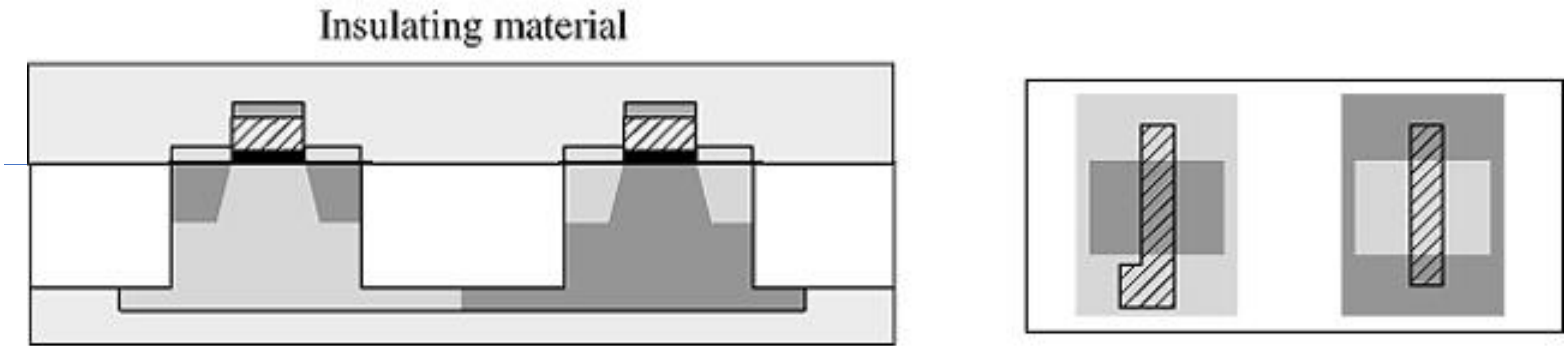


Foundry Masks (Layout) to Physical Devices (5)

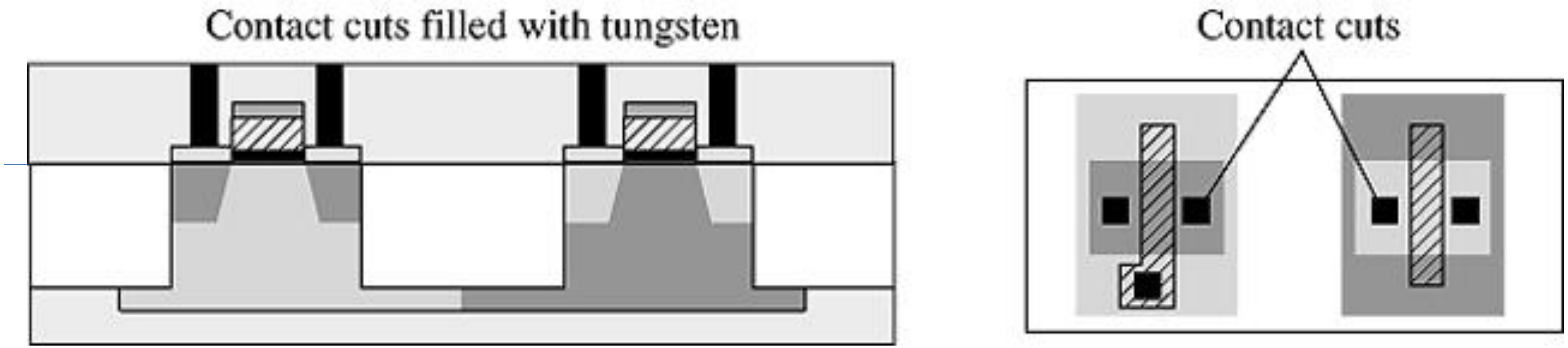




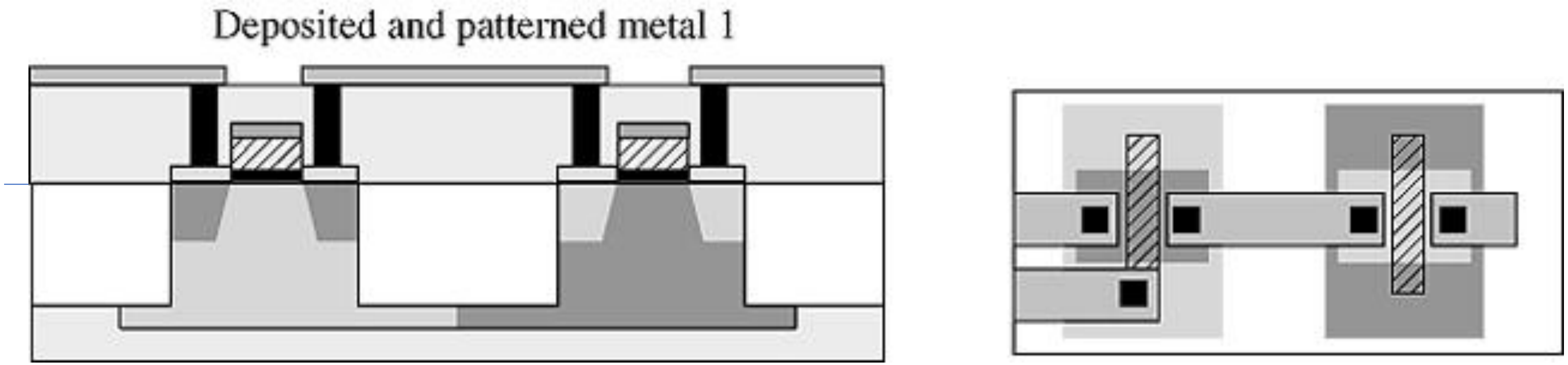
Foundry Masks (Layout) to Physical Devices (6)



Foundry Masks (Layout) to Physical Devices (7)

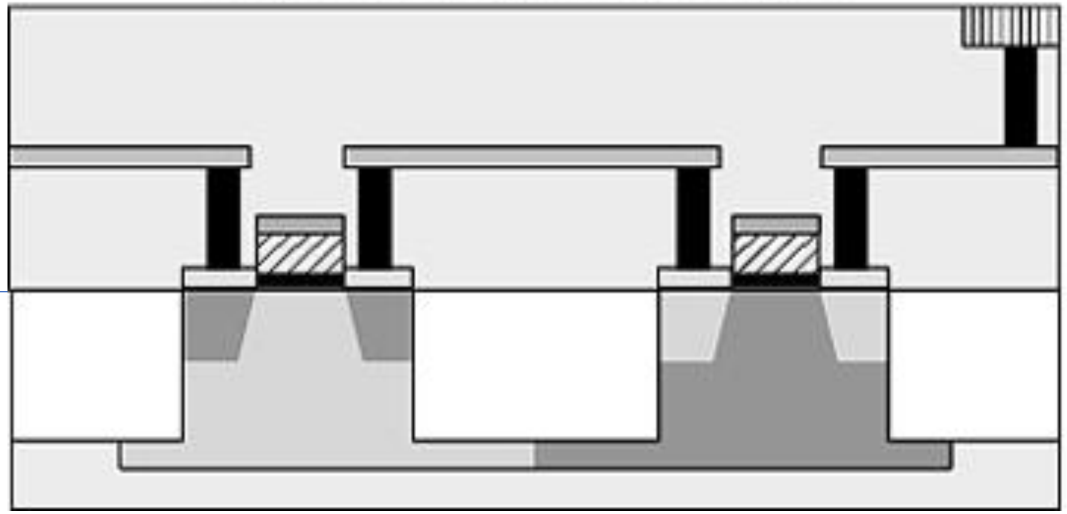


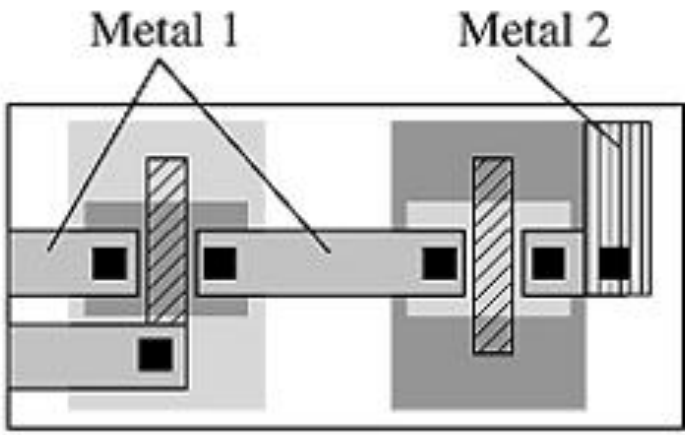
Foundry Masks (Layout) to Physical Devices (8)



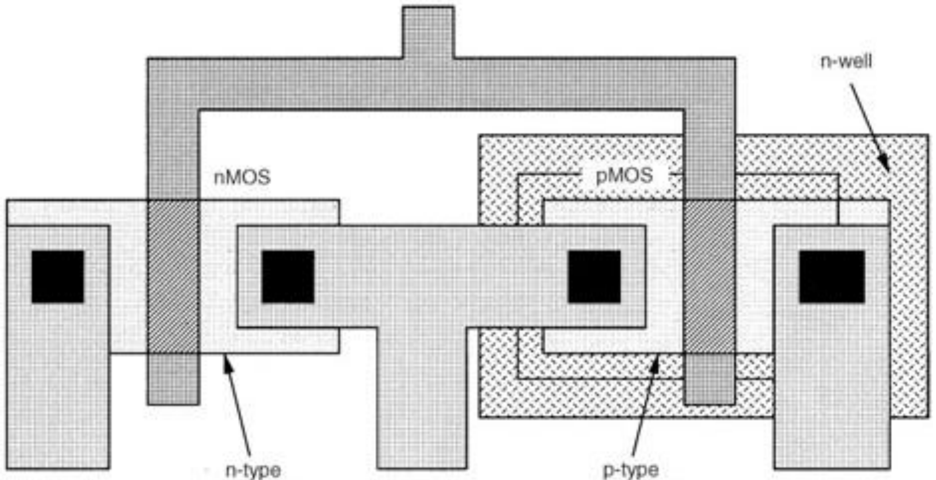
Foundry Masks (Layout) to Physical Devices (9)

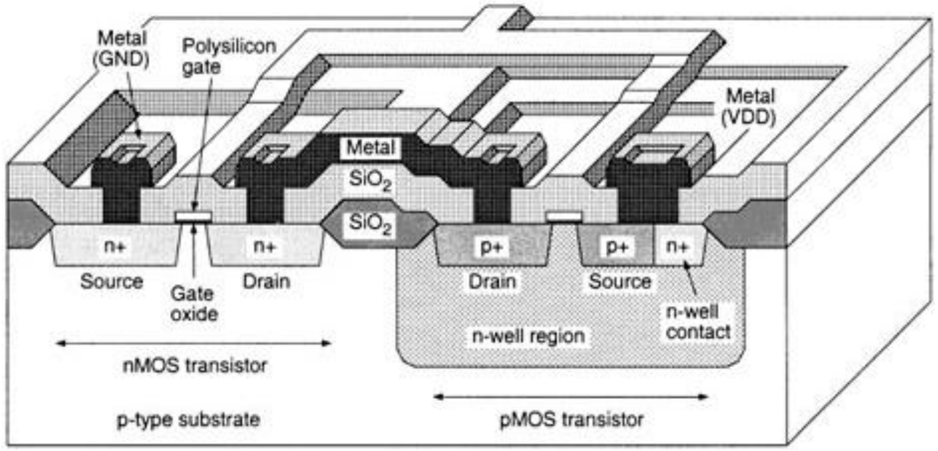
Deposited and patterned metal 2



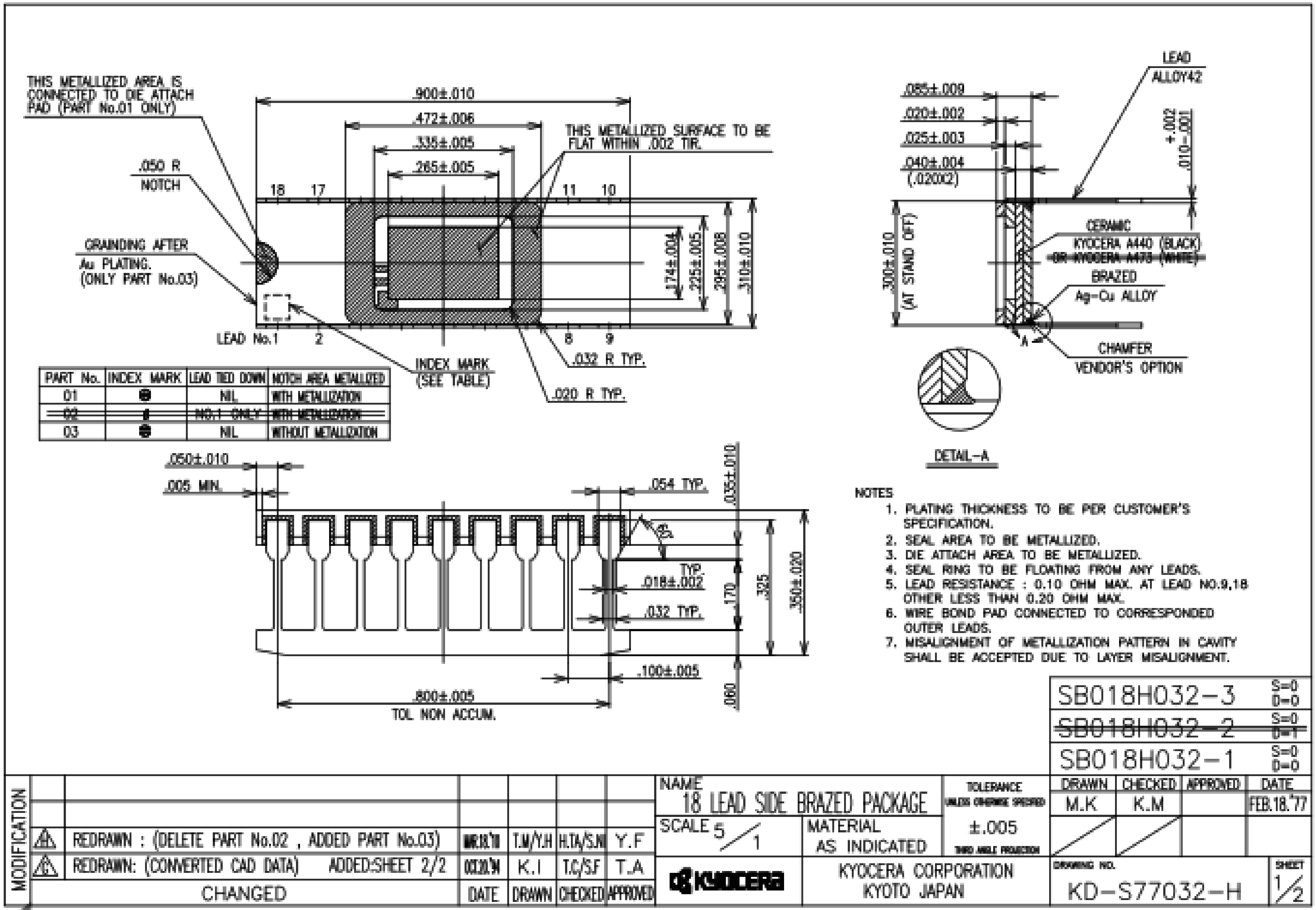


An Older Fabrication Technique: LOCOS

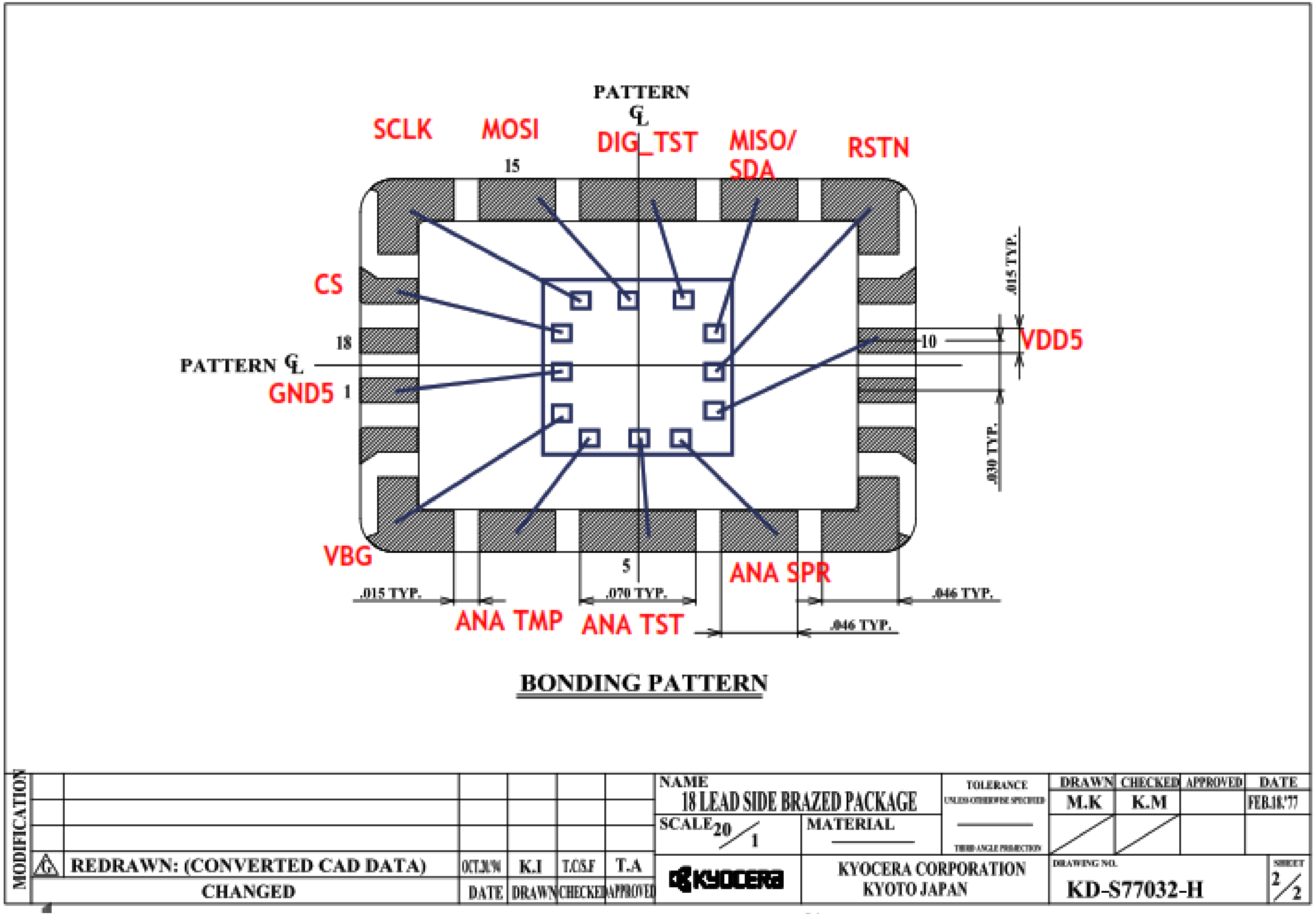




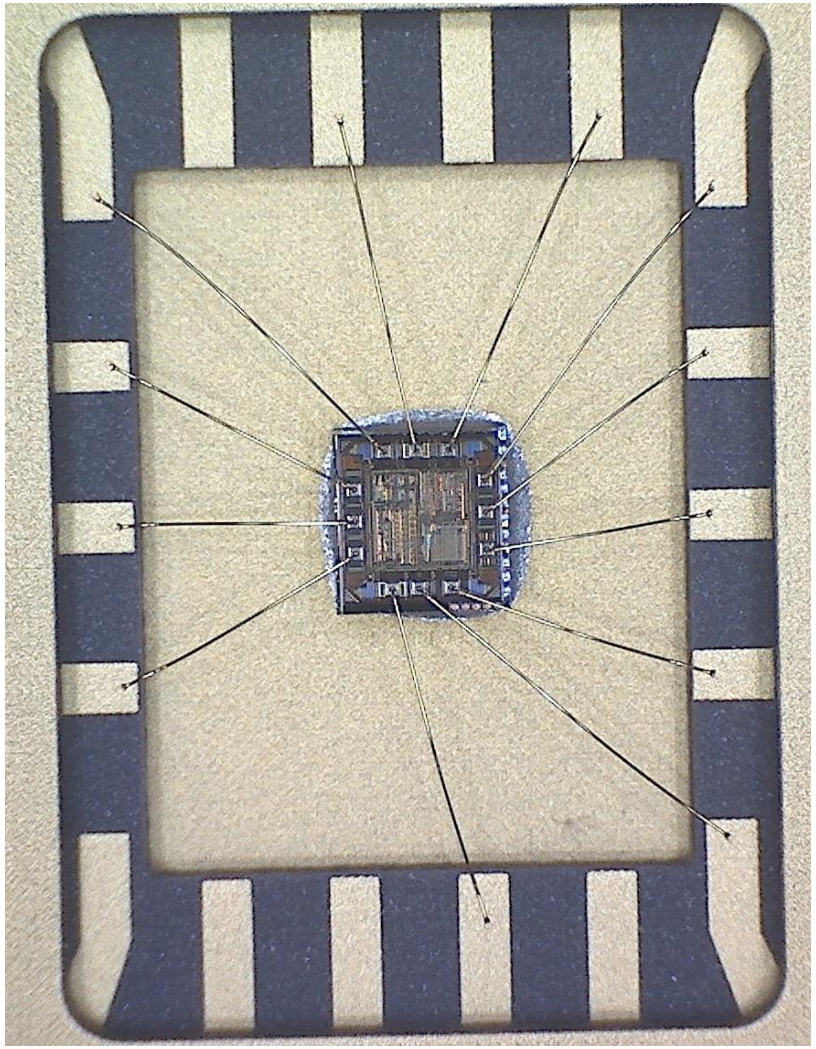
18-PIN CDIP CAD Drawing

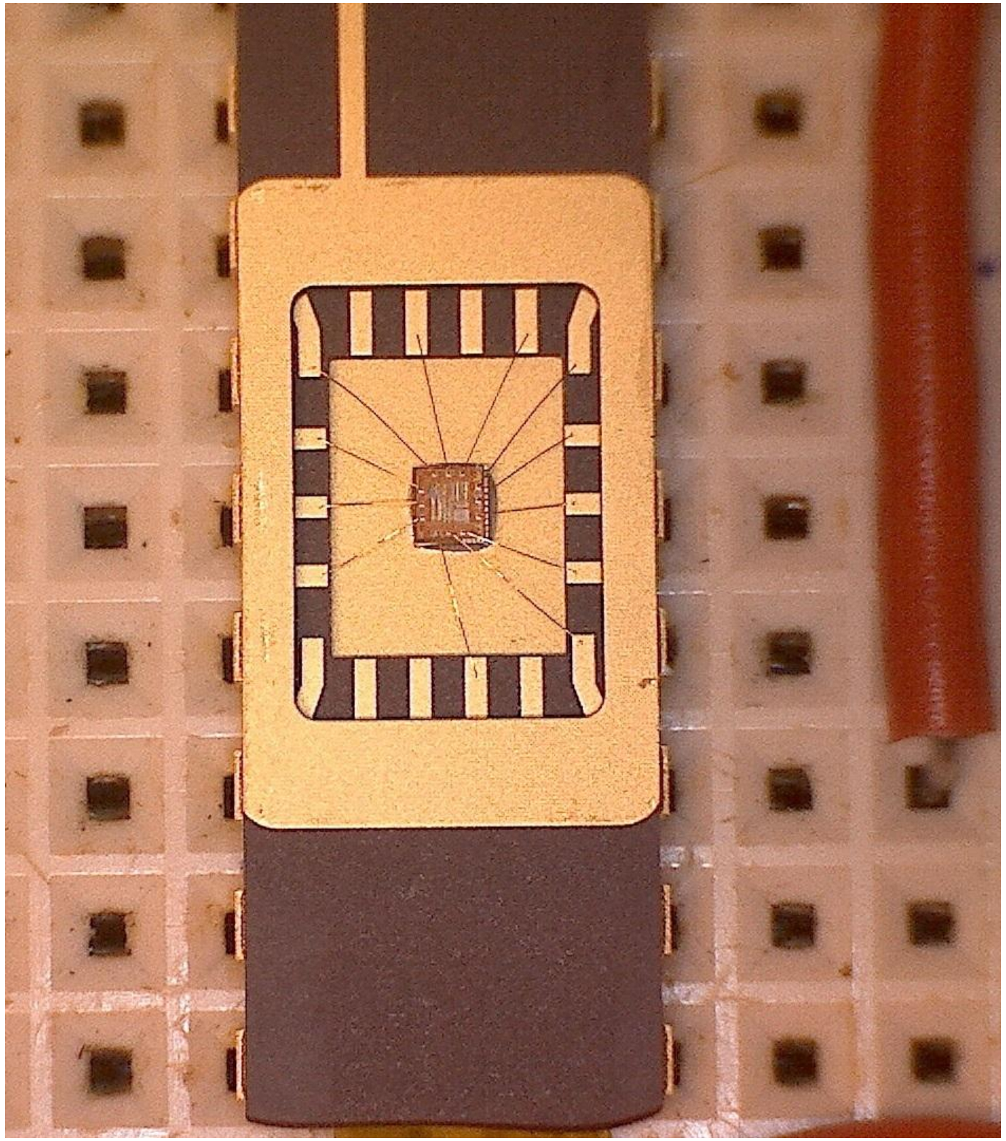


Bonding Diagram

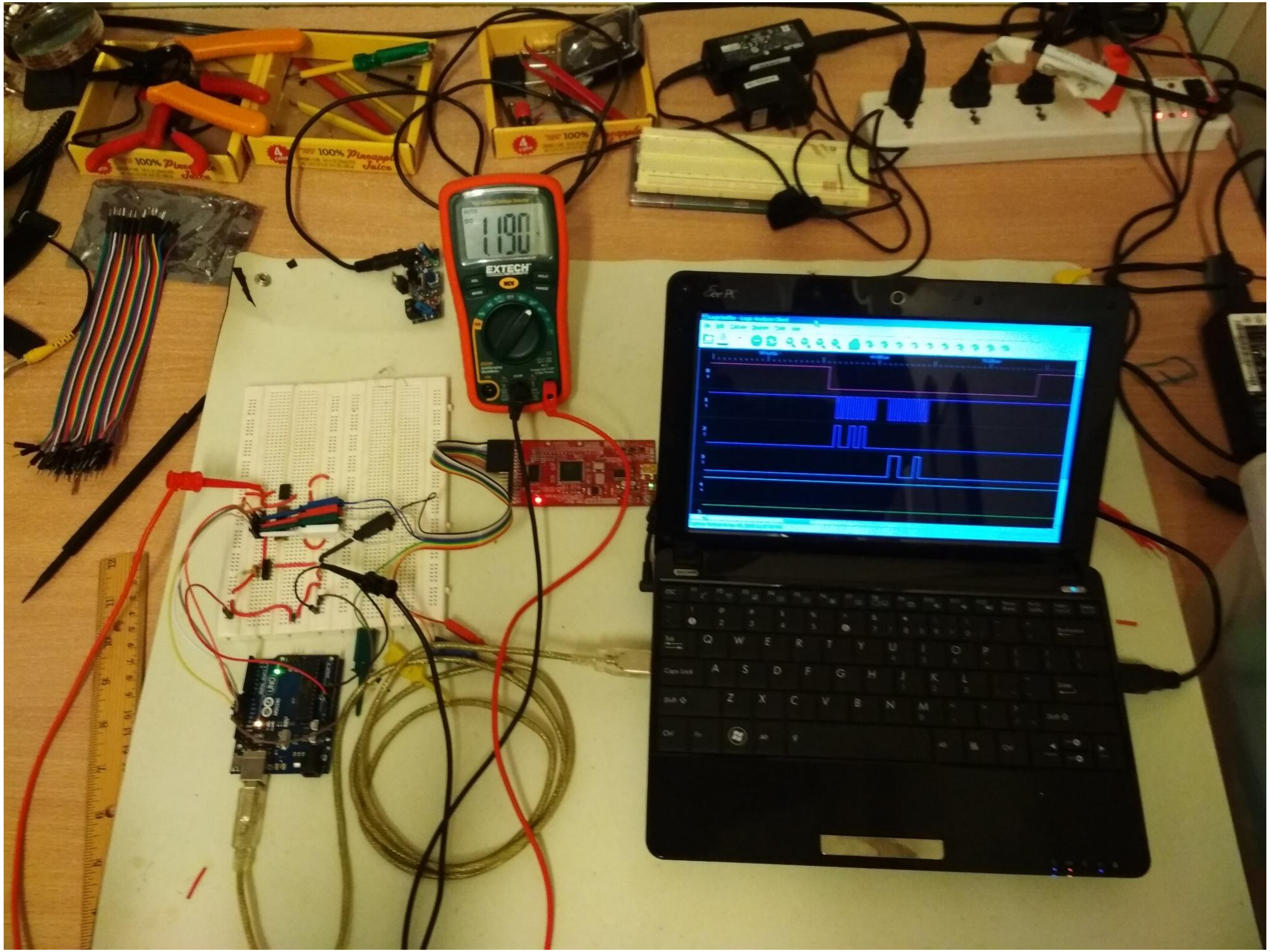


Packaged Part

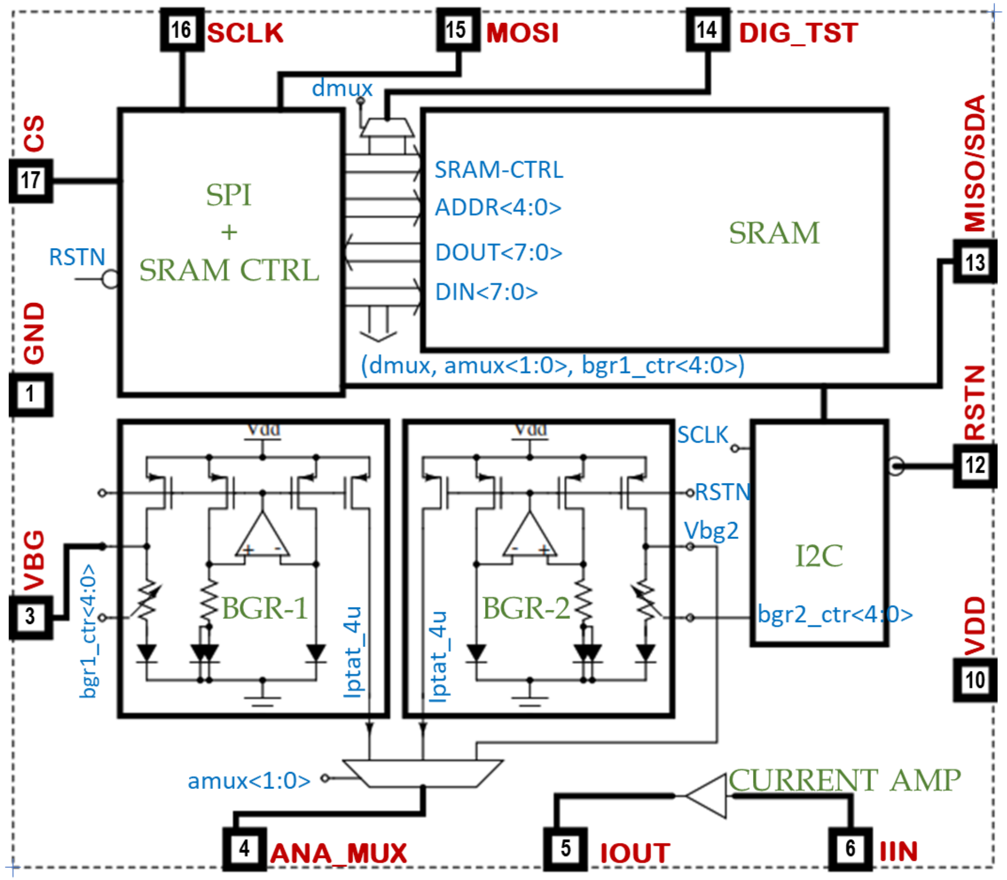


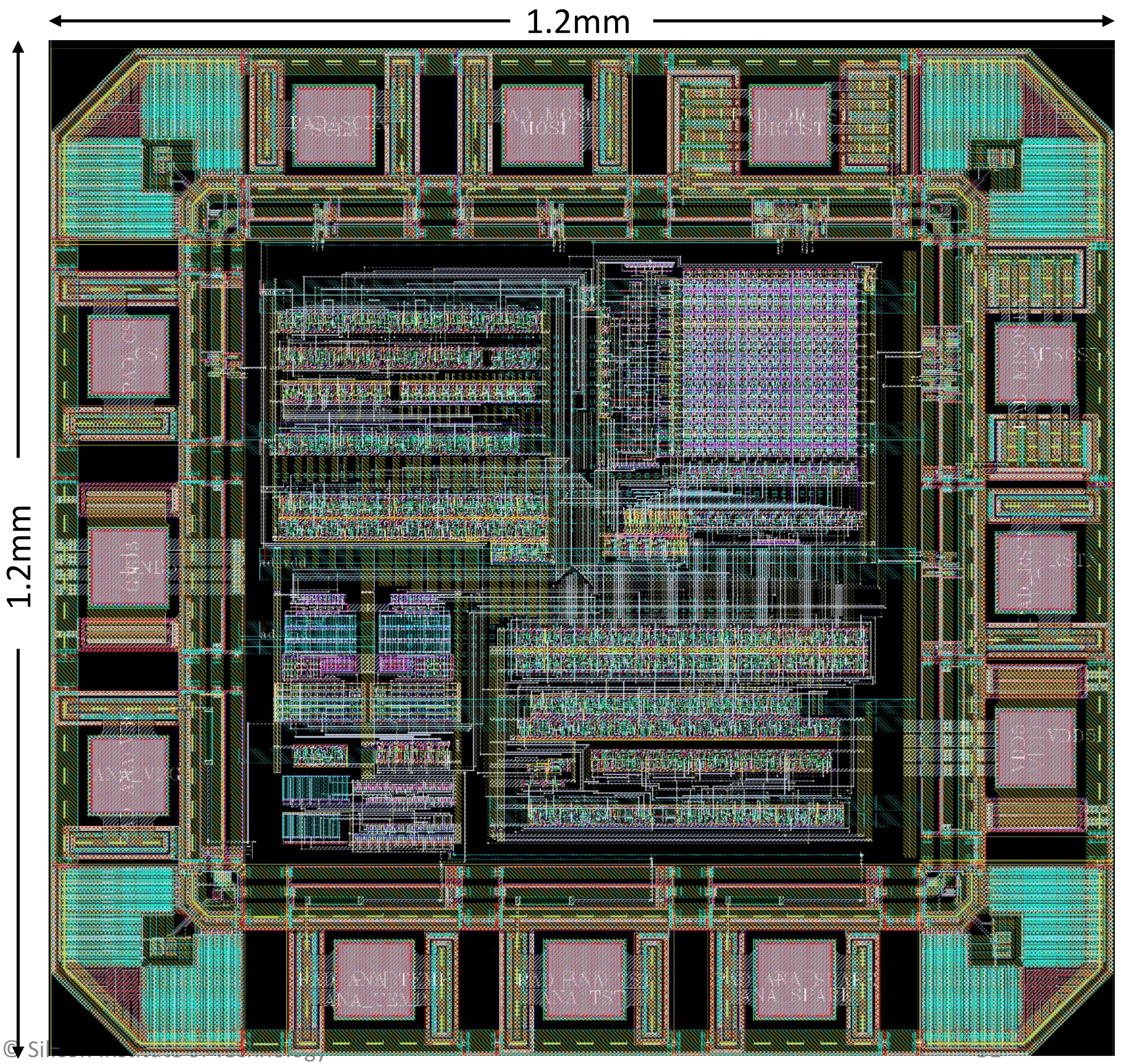


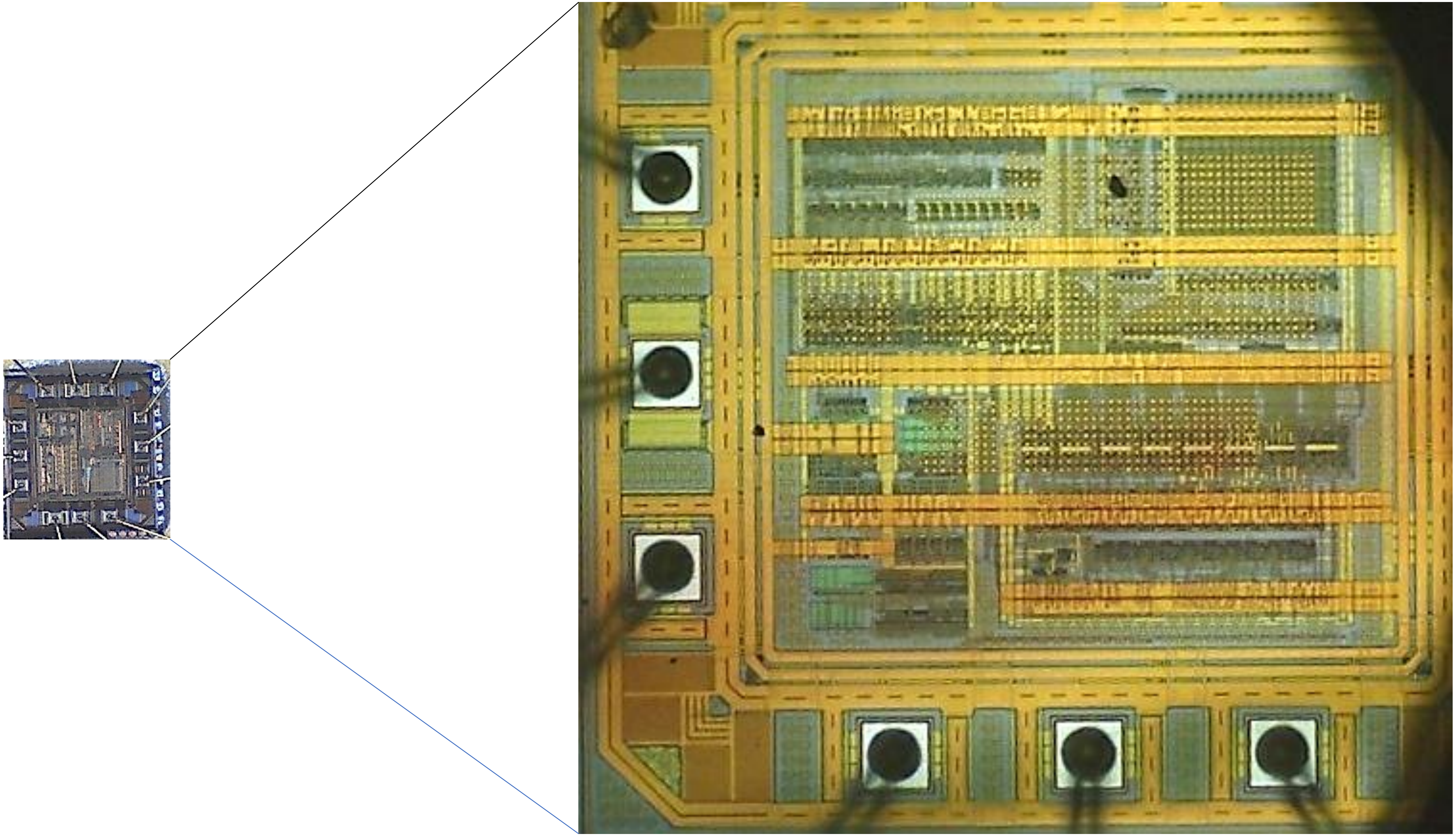
Testing



Project Volta



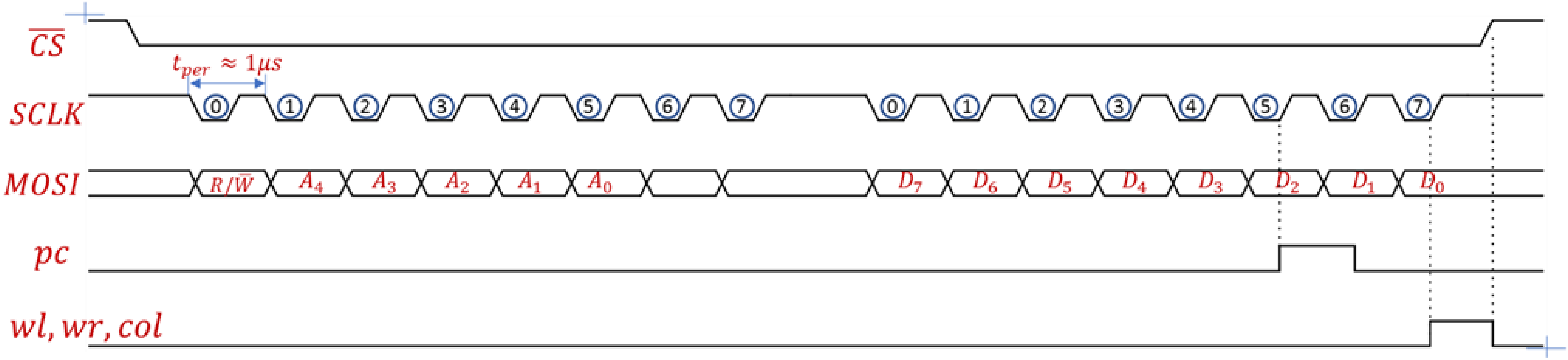




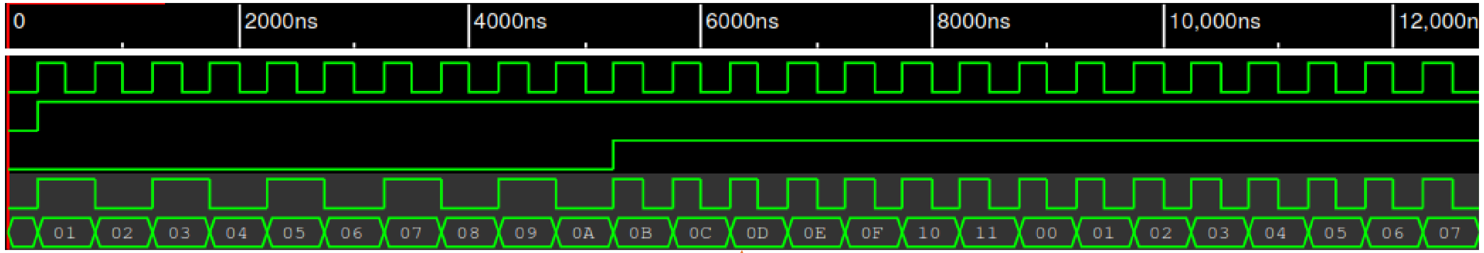
Digital IC Design Flow

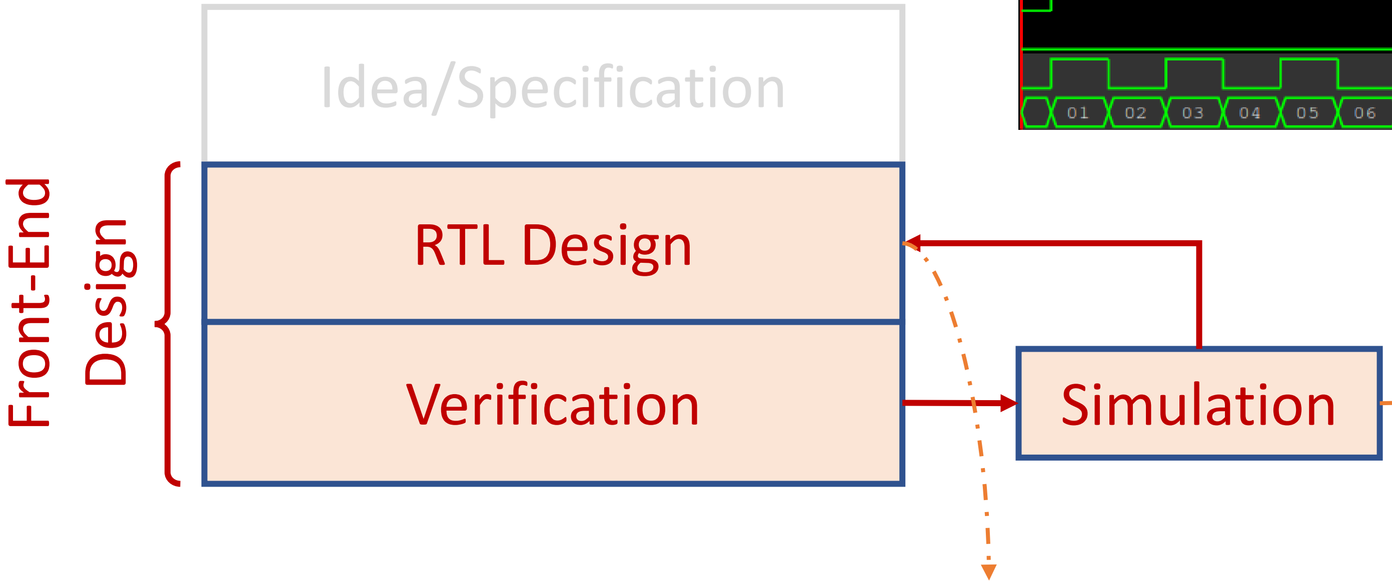
Idea/Specification

Serial Protocol Interface (SPI)



Digital IC Design Flow

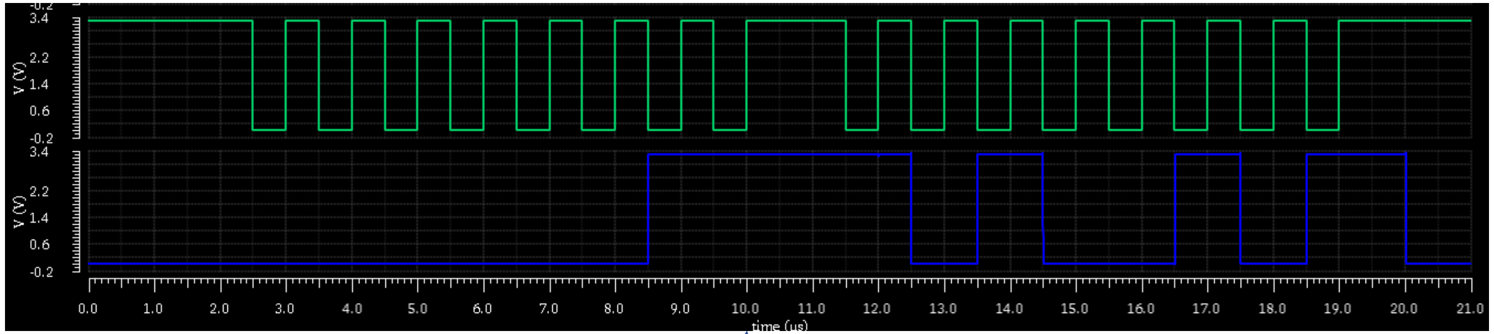





```
// ---------    FLIP-FLOP -----------------
always @(posedge clk_in or negedge rstn_in)
        if (~rstn_in )
                clk_half <= 1'b0;
        else clk_half <= ~clk_half;

//----------  MUX -------------------------
   assign clk_mux_out = clk_sel ? clk_in : clk_half;
```

Digital IC Design Flow



Front-End Design

Back-End Design

Idea/Specification

RTL Design

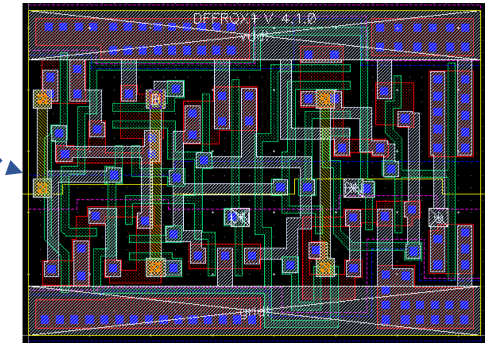
Verification

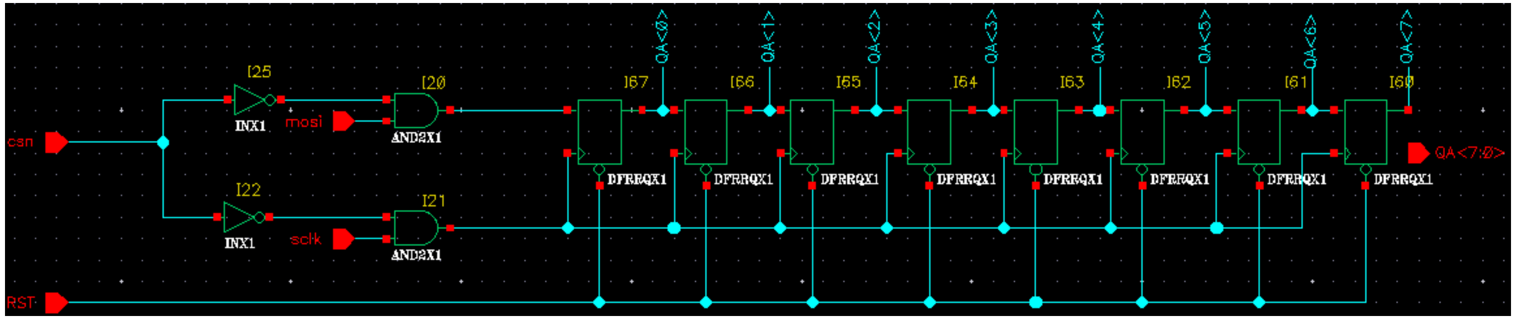
Simulation

Synthesis

Gate Level Sim

Physical Design





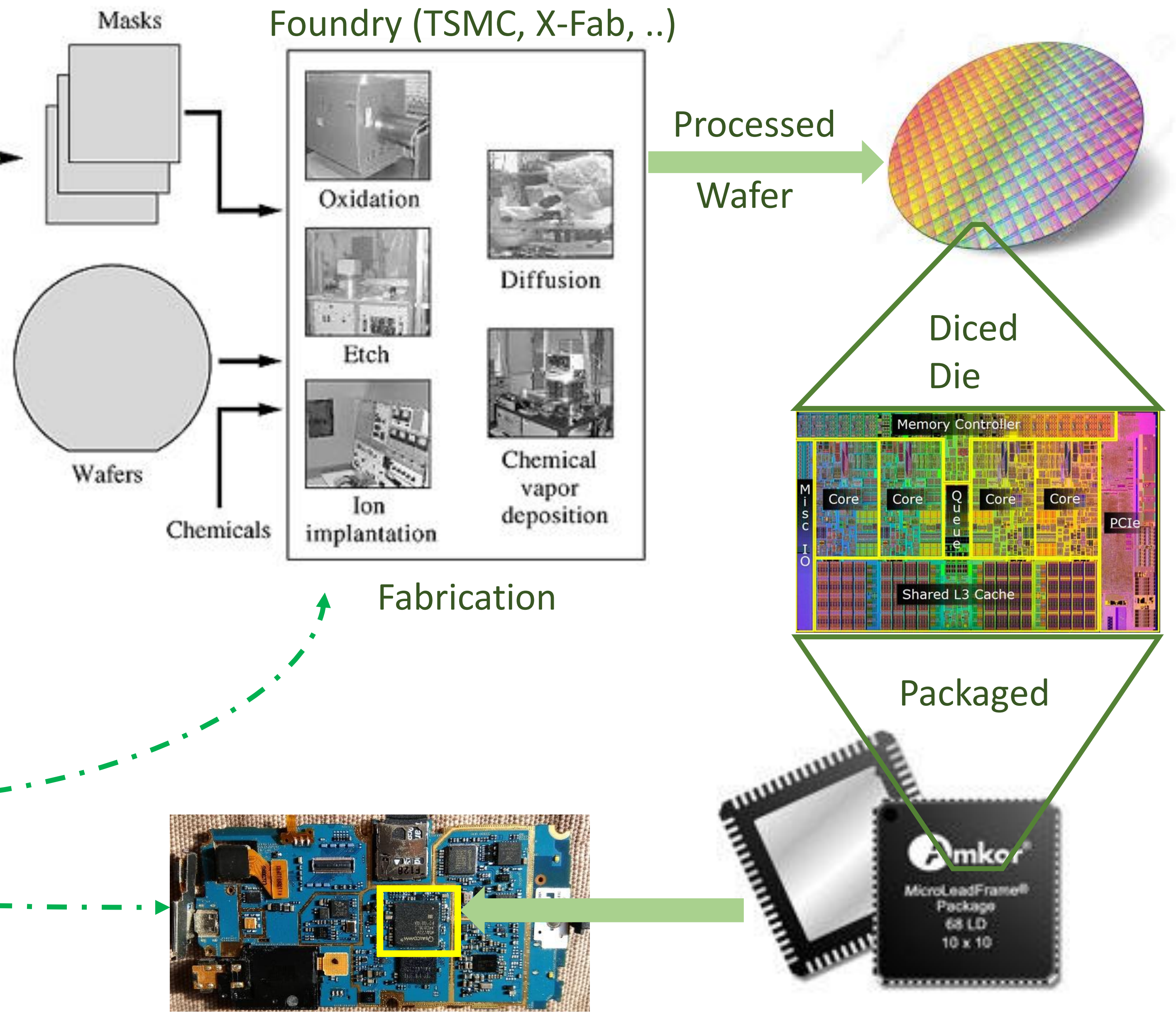
Digital IC Design Flow

Front-End Design

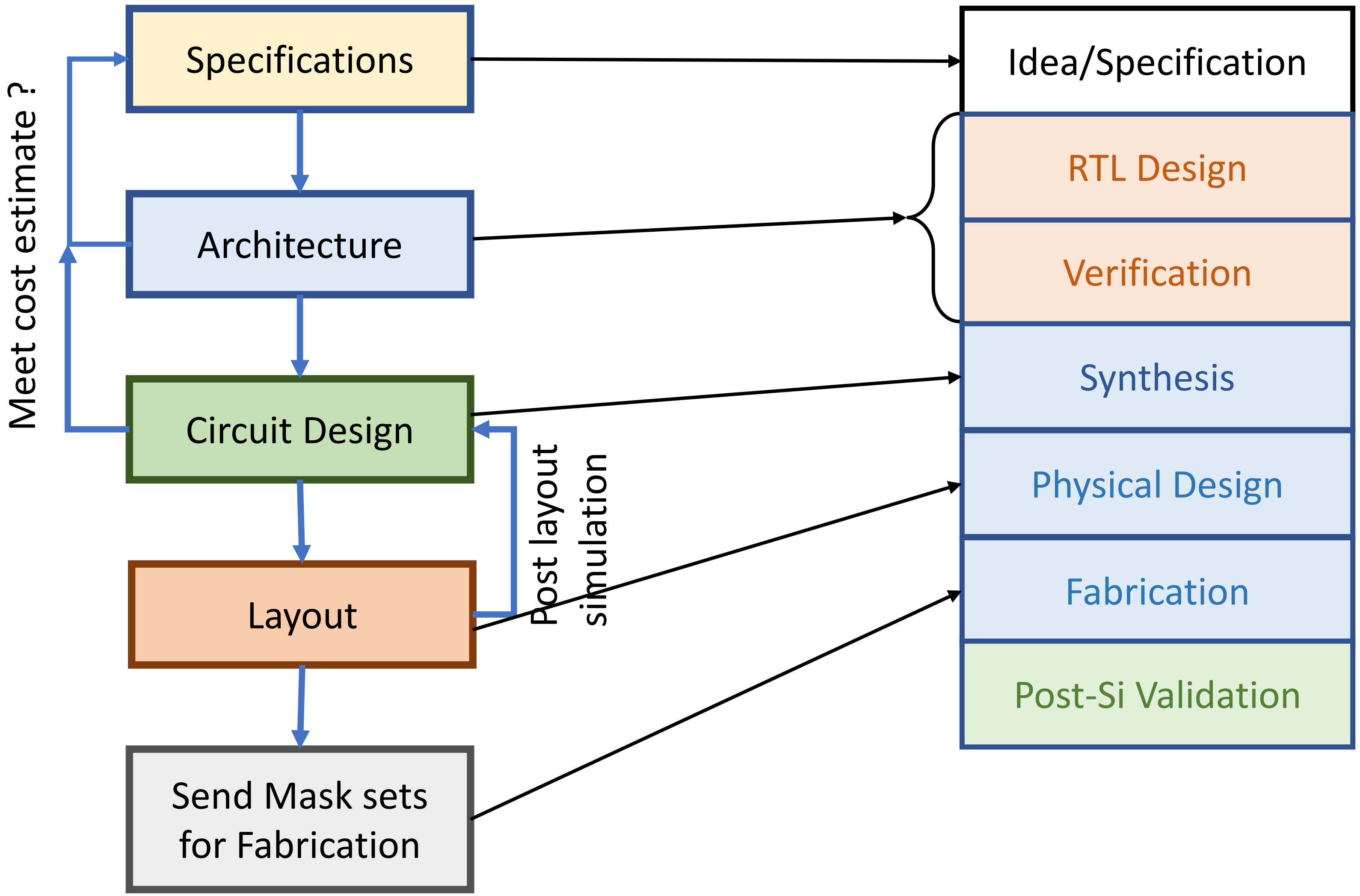
- Idea/Specification
- RTL Design
- Verification

Back-End Design

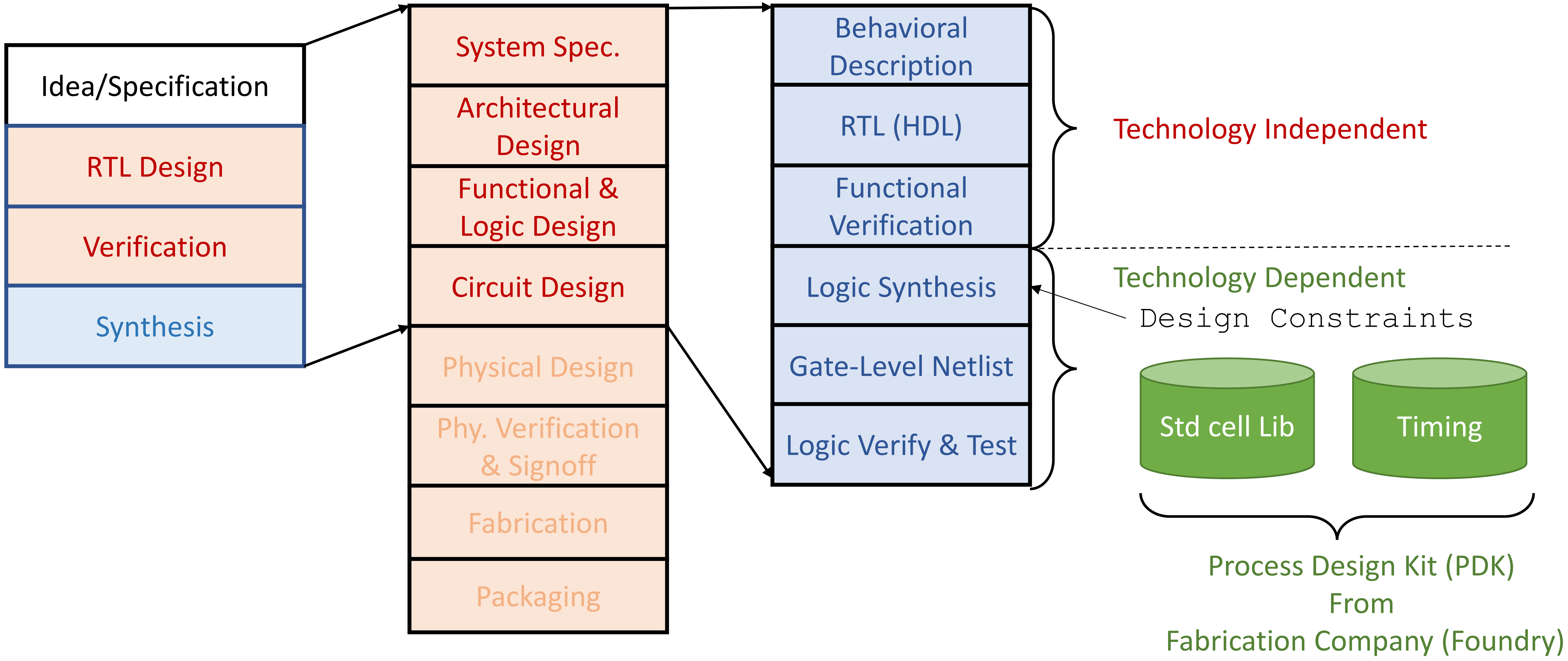
- Synthesis
- Physical Design
- Fabrication
- Post Si Validation



Full Custom vs. Semi-Custom Design Flow



Digital IC Design Flow



Digital IC Design Flow